

Convergence of MEC and DRL in Non-Terrestrial Wireless Networks: Key Innovations, Challenges, and Future Pathways

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Abstract—The rapid growth in mobile communication technologies has turned mobile edge computing (MEC) into a paradigm-shifting technology that extends cloud-like capabilities and storage resources to the edge of the network. This allows computation-intensive and latency-sensitive applications to be performed in close proximity to the end-users, thereby overcoming the bottleneck issues of resource-constrained devices. However, ensuring efficient operations in MEC-empowered systems requires intelligent task execution and resource allocation across MEC servers. To this end, MEC-empowered non-terrestrial wireless networks (MeNT-WiN) systems represent a promising domain where deep reinforcement learning (DRL)

can serve as a powerful tool to enhance the MEC abilities in edge servers and network entities. This paper presents a thorough overview of the applications of DRL in MeNT-WiNs. In particular, it underlines the main contribution of DRL in enhancing the performance of MeNT-WiNs, including autonomous aerial vehicles (AAV) and satellite communications networks. This paper investigates how DRL can meet the unique requirements of MeNT-WiNs by enhancing system efficiency, scalability, and decision-making processes across MEC architectures. First, the article reviews the fundamentals of DRL, it later discusses its integration with MeNT-WiNs and demonstrates its relevance for the optimization of satellite communications and management of AAV swarms, as well as enhancing connectivity in remote areas. The survey also identifies key challenges for DRL-driven MeNT-WiN systems, such as computational complexity and real-time adaptability, while being scalable. Finally, it discusses future research possibilities, emphasizing the importance of new solutions that integrate DRL with MEC in order to fully exploit the potential of MeNT-WiNs.

Index Terms—Mobile edge computing (MEC), MEC-empowered non-terrestrial wireless networks (MeNT-WiNs), deep reinforcement learning (DRL), autonomous aerial vehicles (AAVs).

Received 25 September 2024; revised 23 December 2024, 5 February 2025, 29 April 2025, and 24 May 2025; accepted 26 May 2025. Date of publication 4 June 2025; date of current version 2 January 2026. The work of Haejoon Jung was supported in part by the MSIT, South Korea; in part by the National Research Foundation of Korea under Grant RS-2023-00303757; in part by the ITRC Support Programs under Grant IITP-2025-RS-2021-II212046; and in part by the Convergence Security Core Talent Training Business Support Program supervised by the IITP under Grant IITP-2023-RS-2023-00266615. The work of Aamir Mahmood and Mikael Gidlund was supported in part by KKS Research Profile NIIT, and in part by the EU Interreg Aurora Project ENSURE 6G. (Corresponding authors: Haejoon Jung; Syed Ali Hassan.)

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Digital Object Identifier 10.1109/COMST.2025.3576571

I. INTRODUCTION

ARTIFICIAL intelligence (AI) [1] has witnessed a significant evolution over the past two decades, transitioning from traditional machine learning (ML) frameworks [2] to more sophisticated and powerful models. Early AI systems predominantly relied on conventional ML algorithms, which were designed to discern patterns from data and perform specific tasks without the need for explicit programming. These classical ML models, such as linear regression and decision trees, were highly dependent on handcrafted feature extraction and predefined rules to function effectively. However, as the complexity of real-world tasks increased, these models struggled to capture the intricate structures and latent patterns present in raw data, highlighting their limitations and motivating the shift from ML to its subfield, deep learning (DL) [3]. In DL, the key concept is the use of neural networks (NNs) with numerous layers permitting an automated feature extraction with a hierarchical learning process. This hierarchical learning ability has made DL highly effective in solving complex problems in areas such as image

or speech recognition, natural language processing (NLP), and even mastering Go and chess, all without prior domain knowledge. Because it eliminated the need for handcrafted feature generation at scale, it allowed the creation of much more flexible AI systems on top of problems that could not be solved before due to their complexity.

Building on the advancements in deep learning, reinforcement learning (RL) [4] introduced a novel approach that enables intelligent agents to interact with their environments and learn optimal strategies using the paradigm of trial and error, which enables AI to behave in such a way

as to maximize cumulative rewards over time. Therefore, it achieves great breakthroughs in areas like robotic control and strategic gameplay. Further evolution to deep RL (DRL) widened these opportunities by merging the capability provided by deep NNs with those related to making decisions based on RL scenarios. Such integration of AI models further allowed the solution and acting upon more complex unstructured problems while learning ways of navigating them; this, in turn, will significantly enhance developments in major evolving applications including non-terrestrial networks (NTNs), industrial Internet-of-Things (IIoT), smart cities, NLP, autonomous vehicles (AVs), and healthcare. DRL gives the ability to make complex decision-making autonomous for AI even in highly dynamic or uncertain scenarios. Overall, the evolution from traditional ML to DL and from RL to DRL marks the unabated improvement and extension of the problem-solving capabilities of AI. This continuing development opens up new areas in complex decision-making and improves the capability of AI systems to act more autonomously in a variety of applications.

The evolution of DRL has brought a revolution in many fields, especially in the case of mobile edge computing (MEC)-empowered non-terrestrial wireless networks (MeNT-WiNs), where its capabilities proved transformative. In MeNT-WiNs that include but are not limited to autonomous aerial vehicles (AAVs), satellite-based communication systems, and high-altitude platforms (HAPs), the use of DRL becomes relevant due to its inherently dynamic and large-scale nature. One of the critical uses of DRL in NTNs could be the optimization of autonomous satellite operations by enabling real-time decision-making of satellites for tasks such as orbit adjustment, collision avoidance, and energy-efficient trajectory planning [5]. Consequently, DRL enhances resilience and efficiency in satellite constellations by autonomously assisting with the management of complex tasks, particularly in low Earth orbit (LEO) deployments, which are integral to next-generation 5G and emerging 6G communication systems. In turn, within the NTN network, DRL provides crucial performance regarding the adaptive route planning of flights and aerial surveillance. Such drones often have to fly in highly dynamic and uncertain environments, whether over disaster zones or areas with very sparse infrastructure. DRL gives these platforms real-time learning and adaptation; hence, it optimizes routing and communication links to ensure reliable low-latency data transmission to ground stations. In resource management within MeNT-WiNs, DRL significantly improves spectrum and energy resource allocation for large-scale, decentralized satellite and drone networks. Given the complexity and

diverse communication needs of MeNT-WiNs, DRL's ability to learn from experience and adapt in real-time provides an effective mechanism for optimizing RF spectrum usage, mitigating interference, and ensuring fair resource allocation. In addition to this, MeNT-WiN's cybersecurity has now begun applying DRL since highly sensitive communication channels need to be kept safe. DRL-powered systems are used to predict anomalies, attempt to block cyber-attacks, and assess network vulnerabilities, enhancing the robustness of MeNT-WiNs from threats in a time of increased digital perils. The integration of DRL into MeNT-WiNs increases not only the autonomy and performance of those networks but also opens new perspectives regarding innovative opportunities for remote sensing, environmental monitoring, and worldwide connectivity. With its autonomous resource management capability and adaptation to complex and dynamic environments, DRL surely makes MeNT-WiNs play a key role in shaping future global communications and enabling the next wave of intelligent, interconnected systems.

Traditional optimization methods face significant limitations in dealing with the complexity, scale, and dynamic nature of non-terrestrial environments. These methods often rely on static optimization techniques and predefined rules that face challenges while adapting to the highly variable conditions and diverse requirements of MeNT-WiNs. For example, resource allocation schemes based on traditional optimization algorithms are typically computationally expensive and may not effectively respond to real-time fluctuations in network demand or environmental changes. In addition, such approaches often result in suboptimal spectrum utilization, increased interference, and inequitable distribution of resources among network users, especially in decentralized systems such as satellite constellations and AAV networks. Moreover, conventional security mechanisms in non-terrestrial environments are inadequate to address evolving threats such as cyber-attacks and data breaches because they rely on predefined threat models. Similarly, traditional routing and communication strategies often fail to provide low-latency, reliable connections in dynamic scenarios such as disaster recovery or remote sensing missions, where conditions change rapidly. These limitations necessitate the adoption of more adaptive, intelligent, and scalable solutions, positioning DRL as a transformative tool for overcoming these challenges in MeNT-WiNs.

Generally, DRL serves as a transformative instrument for optimization, adaptation, and intelligent decision-making extending beyond these general application domains. It dynamically helps in the allocation of resources to accomplish energy-efficient content delivery in cloud computing (CC) for optimizing server workloads and lowering power consumption [6]. Similarly, in data centers, DRL enables the effective routing of traffic that can reduce latency while enhancing general operational efficiency [7]. In wireless communication networks, DRL ensures profound improvement in the efficiency of the spectrum, adaptive modulation, and dynamic resource allocation, hence dealing with the random traffic loads and diverse user requirements with minimum interference [8]. Such optimization is imperative and indispensable for meeting the high-performance demands of 5G and 6G systems, as resource starvation challenges persist

TABLE I
COMMONLY USED ABBREVIATIONS

Acronym	Description	Acronym	Description
A2C	Advantage Actor-Critic	A2G	Aerial-to-Ground
A3C	Asynchronous Advantage Actor-Critic	ADQL	Asynchronous Deep Q-Learning
AGIN	Air-Ground Integrated Network	AI	Artificial Intelligence
ANN	Artificial Neural Network	ARSUs	Aerial Roadside Units
AV	Autonomous Vehicle	BS	Base Station
CC	Cloud Computing	CER	Combined Experience Replay
CPU	Central Processing Unit	DBN	Deep Belief Network
DDoS	Distributed Denial-of-Service	DDPG	Deep Deterministic Policy Gradient
Due. DQL	Dueling Deep Q-Learning	DDQN	Double Deep Q-Network
Dist DQL	Distributed Deep Q-Learning	D3QN	Double Due. DQN
DL	Deep Learning	DP	Dynamic Programming
DQLNN	Deep Q-Learning with Noisy Network	DRL	Deep Reinforcement Learning
DRN	Deep Residual Network	DRLRM	DRL-based Resource Management
DT	Digital Twin	EE	Energy Efficiency
ETSI	European Telecommunications Standards Institute	FBVN	Federated Blockchain-empowered Vehicular Network
F-DQL	Federated Deep Q-Learning	FL	Federated Learning
FNN	Fully Connected Neural Network	FOMDP	Fully Observable Markov Decision Process
GAN	Generative Adversarial Network	GD	Gredient Decent
GEO	Geosynchronous Equatorial Orbit	GDM	Generative Diffusion Model
GMM	Gaussian Mixture Model	HAP	High-Altitude Platform
I2A	Imagination-Augmented Agent	IIoT	Industrial Internet-of-things
IoAVs	Internet of Autonomous Vehicles	ISAC	Integrated Sensing and Communication
ISATRNs	Integrated Satellite-Aerial-Terrestrial Relay Networks	ITS	Intelligent Transportation System
KNN	K-Nearest Neighbor	KKT	Karush- Kuhn-Tucker
LEO	Low Earth Orbit	LLMs	Large Language Models
LSTM	Long-Term Short Memory	MADDPG	Multi-Agent DDPG
MADRL	Multi-Agent DRL	MAFRL	Multi-Agent Federated Reinforcement Learning
MAPPO	Multi-Agent Proximal Policy Optimization	MARL	Multi-Agent Reinforce Learning
MARS	Multi-Agent Resource Scheduling	MBMF	Model-based RL with Model-free Fine-tuning
MDP	Markov Decision Process	MCC	Mobile Cloud Computing
MCTS	Monte Carlo Tree Search	MEC	Mobile Edge Computing
MeNT-WiNs	MEC-empowered Non-Terrestrial Wireless Networks	MINLP	Mixed-Integer Nonlinear Programming
MIoT	Maritime Internet-of-Things	ML	Machine Learning
MLP	Multi-Layer Perceptron	mMIMO	massive Multiple-Input–Multiple-Output
mmWave	Millimeter-wave	MRS	Multi-task Resource
NFV	Network Function Virtualization	NLP	Natural Language Processing
NOMA	Non-Orthogonal Multiple Access	NTN	Non-Terrestrial Network
OMADDQL	Optimization-embedding Multi-Agent Due. DQL	PDQL	Prioritized DDL
PER	Prioritized Experience Replay	POMDP	Partially Observable MDP
PPO	Proximal Policy Optimization	QoE	Quality-of-Experience
QoS	Quality-of-Service	RAN	Radio Access Network
RDGP	Recurrent Deep Policy Drident	RIS	Reconfigurable Intelligent Surface

(Continued)

alongside the need for seamless wireless communication. DRL is also useful in the context of multimedia streaming and content delivery networks to optimize content distribution, ensuring low latency and improved quality of service (QoS)

for end-users [9]. The DRL-powered content delivery networks can actively manage bandwidth by learning from network conditions and user preferences to reduce congestion and thus offer a better viewing experience. In cybersecurity, DRL

TABLE I
Continued. COMMONLY USED ABBREVIATIONS

Acronym	Description	Acronym	Description
RL	Reinforcement Learning	SAC	Soft Actor-Critic
SARSA	State Action Reward State Action	SAT-IoT	Satellite-based IoT
SCA	Successive Convex Approximation	SDN	Software Defined Network
SEC	Satellite Edge Computing	STIoT	Space-Terrestrial Integrated Optical Transmission
SVM	Support Vector Machine	TD	Temporal Difference
TD3	Twin Delayed DDPG	TOMRA	Task Offloading and Multidimensional Resource Allocation
TRPO	Trust Region Policy Optimization	UAV	Unmanned Aerial Vehicles
UE	User Equipment	USVs	Unmanned Surface Vehicles
VEC	Vehicular Edge Computing	VM	Virtual Machine
V2V	Vehicle-to-Vehicle	-	-

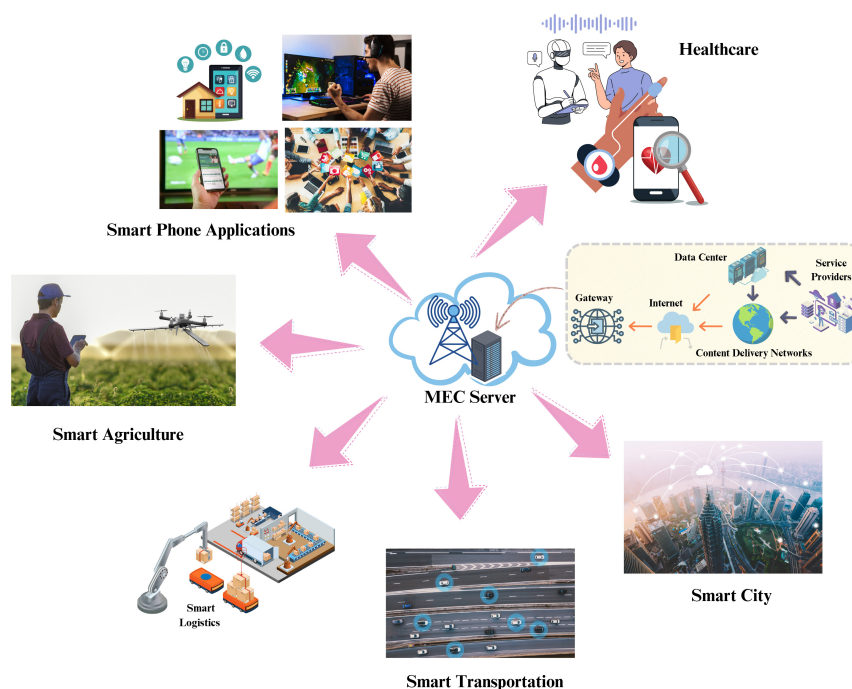


Fig. 1. Illustration of MEC architecture.

detects and identifies network anomalies through learning from data patterns to recognize potential threats and vulnerabilities. It is very valuable in the battle of hardening digital infrastructures since it learns and evolves with emerging cyber threats [10]. In summary, the broad applicability of DRL encompasses various network domains, transforming resource management, communication efficiency, security, and adaptability. By integrating DRL into these areas, organizations can drive innovation, optimize processes, and ensure robust, scalable network performance across diverse applications.

A. CC

CC has emerged as a transformative paradigm in modern computing, providing on-demand access to vast computational resources and storage without the need for direct user management. This paradigm has played a crucial role in supporting the exponential growth of mobile Internet traffic

and the increasing complexity of mobile applications. The advent of mobile CC (MCC) extended these capabilities to mobile devices, enabling remote data processing and application execution at centralized cloud data centers. However, despite its advantages, MCC faces significant challenges in meeting the demands of latency-sensitive applications. The long propagation distances inherent in MCC often lead to excessive latency, which is unsuitable for real-time applications like online gaming, virtual reality, and ultra-high-definition video streaming [11]. Additionally, the rising number of IoT devices and bandwidth-intensive services in 5G networks has further highlighted the limitations of centralized cloud architectures. Consequently, while MCC remains an important component of modern networks, it is insufficient to meet the stringent requirements of next-generation systems. As a result, there is a pressing need for architectural innovations that combine the scalability of CC with the low-latency requirements of emerging applications.

B. MEC

To address the challenges faced by the CC, the European Telecommunications Standards Institute (ETSI) proposed the concept of MEC, also referred to as multi-access edge computing. The ETSI defines MEC as a service environment where CC capabilities are integrated with information technology services at the network edge, especially within close proximity to mobile users and the radio access network (RAN). MEC decentralizes the cloud resources and moves the computing power and data storage closer to the edge of the network, hence creating an ultralow latency and a more reliable system. While MEC is integrated with the RAN, mobile traffic can bypass the traditional core network by routing data through the nearest cloud-enabled edge server. This localized cloud infrastructure reduces bottlenecks, improves base station's (BS's) computing performance, and reduces potential system failures [12], [13]. Through decentralized cloud services, MEC not only improves latency performance, but also expands the range of services available to consumers, enterprises, and industries, making it easier to deploy mission-critical applications. This paradigm shift opens up new value chains, unlocks diverse application scenarios in different industries, and effectively changes the future of 5G and 6G service delivery. Fig. 1 shows the basic architecture of a MEC-enabled system and demonstrates the integration of cloud services at the network edge.

C. CC Versus MEC

There exist notable differences between CC and MEC, as shown in Table II. They differ fundamentally in their architectures, operational paradigms, and application scopes. CC relies on centralized data centers, often located far from end-users, to provide scalable computing and storage resources over the Internet. While this centralized approach supports high-volume data processing and large-scale computing, it introduces significant latency and backhaul traffic due to the long distances between users and cloud servers. In contrast, MEC decentralizes these resources by integrating compute and storage capabilities directly at the network edge, closer to end-users and access points. This proximity enables MEC to deliver ultra-low latency [14], reduce network congestion, and support real-time applications that CC cannot efficiently handle, such as autonomous driving [15], augmented reality [16], and industrial automation [17]. In addition, CC typically requires robust, high-capacity core networks to handle traffic, while MEC mitigates this dependency by offloading tasks locally, resulting in more efficient network utilization. While CC excels at providing global scalability and resource availability, MEC is designed to complement it by addressing the latency-sensitive and localized processing requirements of next-generation networks, such as 5G and 6G. Together, these paradigms create a synergistic architecture that balances scalability, efficiency, and responsiveness for diverse applications.

D. Computation Offloading Models in MEC Systems

Computation offloading is a critical mechanism in MEC systems that enables resource-constrained devices to meet the stringent requirements of modern applications by leveraging

edge servers. Offloading allows mobile devices to transfer computationally intensive tasks to MEC servers, alleviating their limited processing and energy resources. Depending on the characteristics of the applications and their computational demands, offloading strategies can be broadly categorized into two types: binary offloading and partial offloading. These models differ in their task partitioning granularity, complexity, and ability to handle dependencies between task components. In the following sections, we explore these models in detail, highlighting their key features, use cases, and underlying frameworks.

1) *Binary Offloading*: Binary offloading is a computation task model tailored for scenarios where tasks cannot be partitioned and must be executed entirely at a single location. Specifically, such tasks are executed either locally on the mobile device or fully offloaded to a MEC server. The model represents tasks using a three-field notation $X(M; \tau; N)$, which captures the input data size M (in bits), the completion deadline τ (in seconds), and the computational workload N (in central processing unit (CPU) cycles per bit). These parameters characterize the computational and communication demands of applications and facilitate a straightforward evaluation of latency and energy consumption. The task is required to meet a strict deadline, ensuring completion within τ . However, this model can be extended to accommodate soft deadlines, permitting a small fraction of tasks to be completed after τ . In this scenario, the computational workload N is modeled as a random variable. Binary offloading, although simple, is widely used in the literature due to its ability to effectively model highly integrated or atomic tasks while maintaining mathematical tractability.

2) *Partial Offloading*: Partial offloading, in contrast to binary offloading, is designed for applications composed of multiple procedures or components, enabling a fine-grained division of computational tasks. In this approach, a program can be partitioned into two segments: one executed locally on the mobile device and the other offloaded to the MEC server. The simplest representation of partial offloading is the data-partition model, where input data bits are assumed to be independent and divisible into separate groups for parallel execution across different entities in the MEC system.

However, real-world applications often exhibit dependencies among procedures, significantly influencing the execution and offloading process. For instance, the outputs of certain components may serve as inputs for others, mandating a particular order of execution. Additionally, some functions may be restricted to local execution due to software or hardware constraints, such as image display routines. To address these complexities, a more sophisticated task model called the task-call graph, is employed, which is represented as a directed acyclic graph $K(V; D)$, the vertices V denote distinct procedures, while the edges D denote their dependencies. The task-call graph supports three typical dependency models—sequential, parallel, and general—and accommodates detailed specifications for computational workloads, resource requirements, and input/output data for each procedure. This model effectively captures the interdependencies among components and provides a robust framework for optimizing execution in

TABLE II
COMPARISON OF CC AND MEC

Aspect	CC	MEC
Server Hardware	Large-scale data centers with numerous high-performance servers.	Small-scale data centers with moderate resources.
Server Location	Deployed in dedicated facilities with sizes comparable to several football fields.	Co-located with wireless gateways, Wi-Fi routers, and 5G/6G BSs
Deployment	Requires complex planning and managed by information technology giants like Google and Amazon.	Managed by telecom operators, enterprises, MEC vendors, and home users; requires lightweight configurations.
Proximity to End-users	Far from end-users, potentially spanning national borders.	Close proximity, typically tens to hundreds of meters.
Backhaul Utilization	Frequent usage, often contributing to backhaul congestion.	Minimal usage, reducing congestion.
System Management	Centralized management.	Employs hierarchical control, with a mix of centralized and distributed architectures.
Latency Support	Generally supports latency above 100 milliseconds.	Capable of achieving approximately 10 millisecond latency.
Applications	Suited for latency-tolerant and compute-intensive services like social networking and e-learning.	Ideal for latency-sensitive and compute-intensive tasks such as autonomous driving, and interactive gaming, and augmented reality.

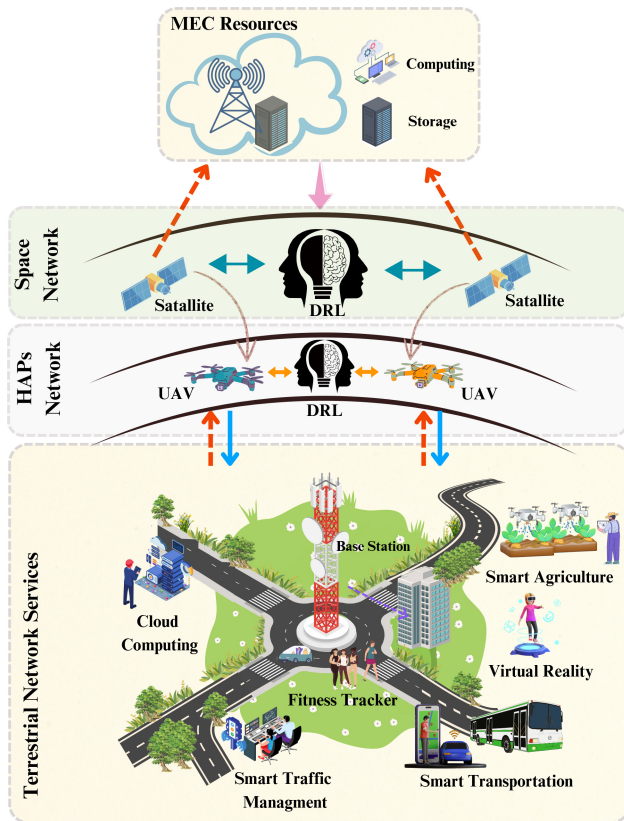


Fig. 2. Illustration of MeNT-WiN.

MEC systems, especially for latency-sensitive and resource-constrained applications [18].

E. MeNT-WiN Architecture

Generally, MeNT-WiNs represent the seamless integration of MEC with NTN platforms, including satellites (e.g., LEO and geosynchronous equatorial orbit (GEO)), HAPs, and AAVs. This architecture facilitates distributed computing and low-latency processing, bringing computational and storage capabilities closer to end-users, even in remote or underserved

areas. The MeNT-WiN architecture comprises several key components. Non-terrestrial platforms act as mobile edge nodes, hosting MEC servers to perform local computation, caching, and analytics. Ground stations and the core network provide backhaul support and coordinate with terrestrial infrastructure, while user equipment (UE), such as IoT devices, mobile sensors, and smartphones, interfaces with the network. Critical features of this architecture include real-time resource allocation for efficient spectrum and power utilization, task offloading from resource-constrained UEs to MEC-enabled platforms, and adaptive network reconfiguration to maintain performance under dynamic conditions such as mobility, interference, and varying link quality. MeNT-WiNs are poised to play a key role in next-generation wireless networks, with immense potential for applications such as disaster recovery, environmental monitoring, remote sensing, and enabling global connectivity. Fig. 2 presents an illustration of a MEC-empowered NTN environment.

The rapid evolution of wireless networks toward 6G systems demands transformative architectures capable of delivering ultra-reliable, low-latency, and ubiquitous connectivity. Traditional terrestrial infrastructures often face significant limitations in remote, rural, or disaster-prone areas due to coverage gaps, backhaul constraints, and scalability issues. In this context, MeNT-WiNs have emerged as a promising paradigm to overcome these challenges by integrating MEC capabilities with NTNs, such as satellites, HAPs, and AAVs. This integration facilitates distributed intelligence, proximity-based computation, and agile service provisioning. By enabling edge processing at non-terrestrial platforms, MeNT-WiNs significantly reduce communication latency, alleviate core network congestion, and support real-time decision-making for delay-sensitive and bandwidth-intensive applications. Furthermore, MeNT-WiNs are envisioned as foundational enablers for a wide array of emerging use cases, including global IoT connectivity, disaster recovery, precision agriculture, remote sensing, and environmental monitoring. These capabilities make MeNT-WiNs a critical component in shaping the future of sustainable, inclusive, and intelligent wireless ecosystems.

F. Challenges Associated With MeNT-WiNs

While NTN assisted by MEC (or MeNT-WiNs) are promising technologies for the next generations of 6G communications, they also face certain challenges owing to specific mobility and propagation scenarios [19], [20]. Evidently, one of the greatest needs is the optimal spectrum-sharing strategy that should prevent interference between terrestrial and non-terrestrial users, especially in cases of overlapping frequency bands. Moreover, with the integration of MeNT-WiNs into terrestrial systems, efficient management of shared spectrum is essential in order to maintain network performance and avoid interference issues. Furthermore, the significant path loss in MeNT-WiNs, especially for satellite and HAP communications, makes higher power demand inevitable to ensure reliable signal transmission. This also poses a serious concern on EE [21] issues, particularly when satellites and AAVs have power-constrained devices.

Similarly, other challenges imposed on MeNT-WiNs include latency [22], resource constraints [23], and variability within networks [24], [25]. High latency is inherent in NTNs, especially for satellite-based systems such as LEO satellites and HAPs, as the distances are long. Dynamic network topologies from mobile platforms like LEO satellites and HAPs further complicate service reliability. In these platforms, there are also severe resource limitations, with confined processing power and energy to support computationally intensive MEC applications. Moreover, security and privacy concerns are heightened by the wide geographic reach of NTNs, increasing their vulnerability to eavesdropping and cyberattacks [26]. Furthermore, seamless integration between terrestrial and non-terrestrial systems is essential for service continuity. However, it presents challenges due to differences in communication protocols and infrastructure requirements. This is an important challenge to address to realize the full potential of MeNT-WiNs in applications such as global IoT, smart agriculture, and disaster management.

G. Limitations of Traditional Optimization Methods in MeNT-WiNs

In the following, we discuss the key challenges posed by conventional optimization methods, particularly highlighting their inadequacies in dealing with the dynamic and complex nature of MeNT-WiNs.

With MEC, there exists a good, robust infrastructure to cope with the high computing requirements of contemporary applications. On the other hand, an inherently dynamic and distributed architecture, as in the MEC system, has incurred several intricacies in comparison to the classic optimization techniques, which have had difficulty staying effective in this environment. Classic approaches, such as convex optimization and heuristics, depend extremely on fixed mathematical frameworks and thus require in advance, static assumptions about the network conditions. While such approaches might suffice in controlled and stable environments, they are generally not rich enough to capture the highly dynamic and heterogeneous nature of MEC ecosystems, especially that of MeNT-WiNs. The unpredictable nature of variability in network topology,

user mobility, and available resources in such scenarios worsens the problems with these traditional methods.

There are quite a few more challenges in MeNT-WiNs due to the implication of rapidly changing non-terrestrial communication environments. Variability in network topology happens quite frequently, and resource availability, as well as user mobility, go up to very high levels with LEO satellites or drones (or HAPs), making it completely inappropriate to rely on classical optimization techniques alone to meet the real-time demands. Moreover, most such approaches rely on off-line optimization procedures where the computation is performed with assumed known parameters rather than adapting to evolving conditions on the fly. In addition, classical methods often have scalability problems. Since MeNT-WiNs typically involve a vast number of interconnected devices and nodes, some standard algorithms may break down under increased computational complexity and delay, making them quite inefficient for large-scale deployments. This is complicated by the fact that one must account for links both on the ground and in space while also accounting for latency, bandwidth, and connectivity differences in such environments. Consequently, this poses a critical limitation on these approaches, as they do not handle stochastic or uncertain environments effectively. MeNT-WiNs are highly sensitive to environmental contexts, like weather conditions, which can very easily degrade the quality of the signal as well as its connectivity. These methods do not offer the degrees of adaptation to uncertainties that would be needed without much re-calibration since they are based on classic optimization algorithms designed for deterministic environments.

H. Role of DRL in MeNT-WiNs

This subsection presents the pivotal role of DRL in MeNT-WiNs. Generally, for a resource allocation scenario, real-time management is the central issue of wireless communication systems. The traditional approach to optimization usually falls short in its ability to adapt quickly to changes in the demand of users, availability of resources, or traffic loads in the network, and therefore, an optimal performance could be suboptimal. For example, the resource allocation practices based on historical trends might have caused inefficiencies where there was an unexpected change in the network conditions. In contrast to this, DRL techniques are excellent in dynamic environments as they exploit their interaction ability with the environment in real-time at all times, as DRL methods support adaptive learning, that support continuous optimization of resources [27], [28], [29], [30]. Specifically, in the case of MeNT-WiNs, arise optimization problems due to user mobility, long-distance transmission, and time-varying propagation conditions for which traditional techniques lack tractability.

DRL when integrated with MEC will make for a formidable solution for the difficulties of the NT-WiNs as this will then provide real-time dynamic resource allocation, spectrum sharing, and interference mitigation between terrestrial and non-terrestrial users. Because DRL is an always-learning mechanism, adjustments can be made in real-time regarding

changes in environments hence enabling faster data transfer within the network to reduce latency. As an illustration, the DRL can optimize handovers of satellites along with routing data on AAVs for connectivity to be established in rural localities, while the DRL can maintain computational loads on edge satellites and ground stations to process data more quickly. The DRL will therefore anticipate how the network would behave and adapt based on changes in conditions for designing energy-efficient communication protocols to meet the energies at the node level. This will also enable DRL to shape optimal spectrum-sharing strategies that can not only mitigate interference but also ensure there is harmonious coexistence between these users on both the ground and in the air without compromising network performance.

Furthermore, by employing DRL in MeNT-WiNs, energy efficiency (EE) can be optimized by intelligent power transmission and optimization of resources, depending on current network conditions. Besides, DRL can guarantee QoS through proactive adaptation of the network configuration to satisfy multi-user and multi-application requirements. Whether bandwidth optimization, interference mitigation, or even load balancing on the network, DRL shall ensure that MeNT-WiNs meet the high performance and reliability standards that are expected within communication networks at 6G [40]. Overall, DRL dramatically enhances the reliability and performance of MeNT-WiNs, letting these networks autonomously adapt to dynamic environments, optimize resource utilization, and preserve robust communication links. The integration of DRL in MeNT-WiNs is a transformational approach that fuels the development of non-terrestrial communication systems toward more efficient, resilient, and intelligent systems.

Building on these capabilities, multi-agent DRL (MADRL) offers an advanced framework for managing the complexities of MeNT-WiNs, where non-terrestrial platforms such as satellites, AAVs, and HAPs often operate in collaborative or competitive scenarios. MADRL facilitates distributed decision-making among multiple agents, enabling them to learn, adapt, and interact simultaneously while considering the strategies and actions of others. For instance, MADRL can optimize the coordination of satellite constellations for dynamic spectrum allocation [41] or enhance AAV trajectory planning in disaster response missions. By fostering cooperation and competition among agents, MADRL addresses scalability and complexity challenges in large-scale, multi-agent non-terrestrial systems, further driving the evolution of MeNT-WiNs into efficient, resilient, and intelligent communication networks.

Meanwhile, hierarchical RL [42] builds upon the successes of traditional RL algorithms, which have achieved remarkable results in various domains including MeNT-WiNs. However, as tasks grow in complexity, the limitations of flat RL approaches become apparent, such as the difficulty of handling long-horizon dependencies and high-dimensional decision spaces. Hierarchical RL addresses these challenges by introducing a structured framework that decomposes intricate tasks into a hierarchy of manageable subtasks. This hierarchical approach enables the system to operate at multiple levels of abstraction, with high-level policies overseeing subtask selection and

low-level policies executing specific actions. By leveraging this structure, hierarchical RL enhances learning efficiency and scalability, making it a powerful tool for tackling complex, multi-layered problems in dynamic environments. This structured approach not only enhances the scalability of decision-making processes in large-scale MeNT-WiNs but also improves resource utilization and system adaptability in highly variable environments [43], [44]. By enabling efficient task decomposition and execution, hierarchical RL aligns seamlessly with the operational demands of MeNT-WiNs, making it a critical tool for optimizing the performance of MeNT-WiNs.

I. Different Perspectives of DRL-Empowered MEC

The rapid growth of IoT devices connected over the Internet has led to the generation of massive datasets, including various media types such as audio, images, and video. Given that this data is produced at the edge of the network, it is advantageous to analyze it locally. Techniques like ML, DL, RL, and DRL are essential for efficient data analysis, quick feature extraction, and learning patterns. Moreover, proper scheduling and offloading tasks to edge computing resources are crucial for effective data processing, ensuring specific data requirements like quality-of-experience (QoE), privacy, and low latency are met. DRL techniques offer robust data prediction capabilities, making them suitable for accurately forecasting data requirements and determining which MEC nodes are best suited for processing. Below, we explore various use cases of DRL-enhanced MEC from multiple perspectives.

1) *DRL-Empowered MEC From Service Providers Perspective:* Bringing DRL models to the edge of the network offers several advantages for stakeholders and service providers. Processing data at the edge, where consumers or nodes generate it, reduces latency and minimizes the costs of transmitting it to remote cloud servers for analysis. This approach is beneficial across various domains such as safety, security, and marketing, including applications like fire monitoring and user segmentation. Moreover, service providers can optimize their long-term profitability by offering computational resources to end-users. For example, in [31], the authors introduced a public blockchain application within MEC, aimed at maximizing the long-term profit of service providers. In this context, miners offload their proof-of-work tasks to MEC-operated BSs, and RL-aided frameworks allow service providers to dynamically adjust prices per unit hash rate while considering miners' highest affordable price during the mining process. The framework also allows miners to efficiently optimize their strategies.

2) *DRL-Empowered MEC From Networking Perspective:* From a networking standpoint, DRL-enhanced MEC focuses on integrating DRL techniques within MEC systems to optimize networking technologies. These technologies include network slicing [56], [57], network function virtualization (NFV) [58], [59], software-defined networking (SDN) [32], [33], content caching [60], [61], network privacy [62], [63], and mobility management [64], [65].

Network slicing, which decouples the physical network infrastructure into several logical networks [36], allows for more efficient use of resources. For instance, [36] explored a 5G network slicing method using NNs and deep belief networks (DBNs). In this method, attributes such as device type and bandwidth are normalized and optimized using hybrid metaheuristic techniques. This approach improves accuracy in predicting slices like enhanced mobile broadband (eMBB), massive machine type communication (mMTC), etc. The hybrid method demonstrates improved accuracy over traditional techniques in simulations.

In NFV, which is a virtualization technology to manage networking functions in software, the hardware costs are reduced, and efficient service orchestration is ensured [34]. NFV enables functions such as traffic analysis and security threat detection [35], and these services run through sequences of service function chains, or virtual network functions [66]. Although NFV presents several advantages, challenges arise, particularly in resource allocation within MEC systems. Offloading and scheduling service function chains efficiently are crucial, as virtual network function execution requires specific sequences. Several works have explored these challenges, including [67], which provided an overview of resource allocation challenges, and [68], which proposed a DRL-aided strategy for service function chain scheduling.

In SDN, the data plane is decoupled from the control plane, enabling dynamic and centralized network resource management [69], [70]. This improves network scalability and efficiency while simplifying management tasks. The integration of NFV with SDN in MEC networks has been explored in several studies, such as LightMANO [71], which presents a lightweight platform for managing network services across distributed NFV systems. Other research focuses on the convergence of SDN, network slicing, MEC in 5G networks [72], and the fusion of MEC and cloud RAN [73]. The LayBack architecture in [74], facilitates resource sharing across networking technologies, while [67] surveys SDN and NFV relationships. Several other studies [75], [76], [77] discussed the benefits of NFV and SDN within MEC systems, particularly for ultra-low latency applications [78].

3) *DRL-Empowered MEC From End-User Perspective:* From an end-user perspective, integrating DRL into MEC systems offers significant advancements in meeting user needs. DRL at the edge maximizes QoE by predicting user requirements. For instance, Facebook developed an ML inference strategy for smartphones to minimize latency and improve QoE [37]. Similarly, LiveDeep uses DL to predict the user's viewport in live virtual reality streaming [38]. In the healthcare domain, HealthFog uses DL on edge devices to analyze heart disease, providing healthcare as a service through IoT devices [39]. Storing raw data on edge devices instead of the cloud enhances privacy while providing real-time data analysis for intelligent decision-making.

DRL techniques also allow end-users to offload latency-sensitive applications, such as online gaming, and mobile augmented reality to nearby edge servers or devices. For example, a DRL-aided approach for task offloading in

mobile augmented reality applications is discussed in [38]. Additionally, a DRL-enhanced joint migration and task-offloading approach proposed in [38] combined long-term short memory (LSTM) and DRL for edge server load prediction, enhancing the accuracy and speed of DRL models during offloading. Table III summarizes the different perspectives of DRL-driven MEC.

J. Related Surveys

Several surveys have been conducted to explore the applications of DRL and other ML techniques across different domains of MEC and wireless communication networks. For instance, the authors in [45] focused on ML-aided proactive caching, emphasizing its role in AAV-aided wireless networks and small cells. Similarly, Luong et al. [46] provided an extensive survey emphasizing the DRL applications in wireless communications and networking, underscoring its potential to optimize network performance and adaptability. In [47], the authors presented a detailed survey linking DL with mobile and wireless networks, offering a comprehensive classification of techniques, platforms, and applications, identifying key challenges, and suggesting future directions for deploying DL in mobile environments. Another survey [48], focused specifically on the applications of DL in edge caching, presenting a taxonomy of DL categories and caching principles for general-purpose networks. Furthermore, in [49], the authors reviewed the computation offloading problems in edge computing, categorizing various approaches, including Markov decision processes (MDPs), game theory, non-convex optimization, Lyapunov optimization, and ML strategies.

Furthermore, the survey in [50] investigated ML-based edge caching, presenting an ML taxonomy and content delivery strategies. Although the authors offered a holistic perspective on the role of ML in cache-aided networks, they allocated limited space to discussing RL-based solutions. Similarly, in [51], the authors conducted an extensive survey on resource scheduling in edge computing, covering techniques such as heuristic methods, approximation algorithms, ML, and game theory. This work was centered on computation offloading, resource allocation, and provisioning, however, it only briefly considered the potential of DRL for these tasks. The survey in [52] delved into DRL-aided caching in heterogeneous networks, however, the study merely presented the complete taxonomy of DRL algorithms and focused primarily on fixed cellular and cooperative networks. Meanwhile, the applications of artificial NNs (ANNs) for wireless network optimization, including edge caching, were explored in [53]. However, the tutorial provided a limited discussion of ANN-aided edge caching and referenced only a small subset of relevant studies. Moreover, the authors in [54] focused on resource allocation in MEC, assisted by ML and DL, however, the authors narrowed their scope to a subset of IoT networks, leaving gaps in the broader exploration of ML and DL's full potential in MEC. Collectively, these surveys reflect the growing body of work dedicated to optimizing MEC and wireless communication systems through ML, DL, and DRL, while also highlighting areas where further exploration, particularly in task offloading

TABLE III
SUMMARY OF DIFFERENT PERSPECTIVES OF MEC

Perspective	Key Focus	Applications
Service Provider	Local data processing to reduce latency and cost, optimization of long-term profitability, data analysis for various domains.	Blockchain applications for MEC; RL frameworks for dynamic pricing and task optimization [31].
Networking	Integration of DRL in networking technologies, optimization of SDN, NFV, network slicing, and privacy.	SDN for dynamic network control [32], [33]; NFV for virtual network functions [34], [35]; network slicing using NNs and deep belief networks [36].
End-User	Enhancing QoE; minimizing latency, privacy, intelligent offloading of applications.	ML inference on smartphones [37]; DL for virtual reality streaming [38]; DL for health analysis [39]; DRL for mobile augmented reality application offloading [38].

and resource management, remains necessary. The surveys conducted in [79], [80], [53], [81] are relevant to networking, however, they are targeting the applications of ML and DL in specific network areas. Furthermore, the survey conducted in [82] discusses the application of DRL in NTN in 6G communication networks. The survey in [55] presented the applications of DRL in MEC-empowered terrestrial networks specifically focusing on terrestrial IoT and vehicular networks.

Despite significant advancements in communication technologies, the aforementioned studies have primarily focused on ML/DL-enhanced IoT or DRL-enhanced communication, with limited exploration of their applications in NTNs. Specifically, prior works rarely explore the potential of integrating MEC with DRL-empowered NTNs and hardly provide a comprehensive analysis of DRL-enhanced NTNs empowered by MEC. Traditional optimization methods used in NTNs face inherent limitations, including their inability to handle the scalability, adaptability, and real-time decision-making demands of these networks. As a result, these methods often lead to suboptimal resource allocation, inefficient spectrum utilization, and inadequate energy management—key challenges that remain unresolved in the context of NTNs. DRL has emerged as a powerful solution to address these limitations by offering unique capabilities for optimizing resource management, spectrum allocation, and security in complex and dynamic environments. Recent advances in DRL have demonstrated its ability to adapt to non-terrestrial settings, optimize multi-objective problems, and make intelligent, real-time decisions, thereby overcoming the challenges posed by traditional approaches. Motivated by these developments, this survey delves into the integration of DRL with MEC-enabled NTNs, with particular emphasis on optimizing resource allocation, enhancing EE, and improving system adaptability.

Unlike previous works that primarily focus on terrestrial-centric applications [55], this survey highlights the unique challenges and opportunities associated with DRL in MeNT-WiNs, such as satellite communications and HAPs. It also addresses a critical gap in the existing literature by exploring innovative approaches to integrate DRL within MeNT-WiNs, particularly how DRL can transform resource management, spectrum efficiency, and energy consumption in non-terrestrial environments. By focusing on these aspects, this survey provides a broader and more comprehensive perspective on DRL applications in MeNT-WiNs. A comparative analysis

of this survey with existing works is provided in Table IV, highlighting its contribution to filling the research gap in DRL-assisted MEC-enabled NTN networks and paving the way for future innovations in next-generation wireless networks.

K. Key Contributions

In this article, we present a state-of-the-art survey focusing specifically on the applications of DRL in MeNT-WiN. This comprehensive survey offers a focused examination of DRL's transformative role in NTNs, filling a critical gap in the existing literature and paving the way for future innovations in this rapidly evolving field. Accordingly, our key contributions are summarized as follows:

- We establish a structured framework that systematically explores the applications of DRL within the context of MEC-empowered NT-WiNs (or MeNT-WiNs), examining its impact on optimizing resource management, communication efficiency, and decision-making processes in satellite communication and AAV-based networks.
- The survey provides an in-depth exploration of the foundational principles of DRL, offering a clear understanding of the essential concepts governing its application in NTNs, particularly in enhancing the adaptability and efficiency of these dynamic, and decentralized environments.
- We thoroughly examine the integration of DRL into MeNT-WiNs, showcasing its effectiveness in optimizing satellite systems, improving AAV network performance, and enabling advanced decision-making in challenging, high-mobility, and high-latency environments characteristic of MeNT-WiNs.
- We present and discuss the unique challenges faced while applying DRL in MeNT-WiNs, including scalability, computational complexity, robust performance, and energy-efficient communication. Additionally, we identify emerging trends and outline future research directions, providing a roadmap for advancing the field of DRL in MeNT-WiNs and guiding future developments in 6G communications and beyond.

L. Paper Organization

The remainder of this paper is organized as follows. Section II explores the fundamentals of AI, including a concise overview of supervised, unsupervised, RL, DL, and DRL. It

TABLE IV
COMPARISON OF THE SURVEY WITH THE EXISTING LITERATURE

Ref.	AI/ML/DL/DRL Taxonomy	DRL-enhanced MEC Use Cases	DRL-enabled MEC	ML-enabled MEC	DL-enabled MEC	Non-Terrestrial Nets.			Main Focus of the Paper
						ML Focused	DL Focused	DRL Focused	
[45], 2019	✗	✗	✗	✓	✗	✓	✗	✗	Explored ML-driven proactive caching strategies within UAV-aided networks and small cell infrastructures to enhance network efficiency.
[46], 2019	✓	✗	✗	✗	✗	✗	✗	✗	Examined the applications of DRL within the domains of communication and networking.
[47], 2019	✓	✗	✗	✓	✓	✗	✗	✗	Explored the integration of DL with mobile and wireless networks, providing insights into techniques, platforms, applications, and future deployment directions.
[48], 2020	✓	✗	✓	✗	✓	✗	✗	✓	Examined the applications of DL in edge caching, with a focus on DL categories and caching principles across various network types.
[49], 2020	✗	✗	✓	✗	✗	✗	✗	✓	Reviewed approaches to computation offloading problems in edge computing, including MDP, game theory, non-convex optimization, Lyapunov optimization, and ML methods.
[50], 2021	✓	✗	✗	✗	✓	✗	✗	✓	Investigated ML-enhanced edge caching, offering a comprehensive ML taxonomy and a detailed comparison of ML solutions in cache-aided networks.
[51], 2021	✗	✗	✗	✓	✗	✗	✗	✗	Reviewed heuristic methods, approximation algorithms, ML strategies, and game theory approaches for scheduling resources in edge computing, encompassing resource provisioning, allocation, and computation offloading.
[52], 2022	✗	✗	✓	✗	✗	✗	✗	✓	Examined DRL-aided caching in heterogeneous networks, with a primary focus on fixed cellular and cooperative networks.
[53], 2019	✗	✗	✗	✗	✓	✗	✓	✗	Explored applications of ANNs for optimizing wireless networks, with a specific focus on edge caching applications.
[54], 2022	✗	✓	✗	✓	✓	✗	✗	✗	Investigated the resource allocation optimization problem in MEC-enabled IoT networks, enhanced by ML and DL techniques.
[55], 2024	✓	✓	✓	✓	✓	✗	✗	✗	Provided a comprehensive taxonomy of AI/ML/RL/DRL, with a focus on the applications of ML, RL, and DRL in MEC-empowered terrestrial IoT and vehicular networks.
This Survey	✓	✓	✓	✓	✓	✓	✓	✓	Provides a comprehensive taxonomy of AI/ML/RL/DRL, with a focus on the applications of ML, RL, and DRL in MeNT-WiNs.

also presents a classification of DRL models, with a focus on value-based, policy-based, and actor-critic methods. Section III investigates the applications of DRL in NTNS, including AAV-assisted systems, and satellite–terrestrial integrated networks. Specifically, it explores DRL-driven solutions in MeNT-WiNs for resource management, EE, service continuity, task offloading, and ensuring security and privacy across diverse MeNT-WiN environments. Section IV presents key challenges, unresolved issues, and potential future research directions. Finally, Section V concludes the paper by summarizing the main insights and contributions. An overview of the organization of the paper is provided in Fig. 3.

In summary, this section introduced the foundational concepts of AI and traced its evolution from ML to DRL. We provided an overview of MEC and examined its key components, followed by the motivation for integrating MEC with NTNs to establish MeNT-WiNs. The discussion highlighted the inherent limitations of purely terrestrial infrastructure and outlined the challenges associated with traditional optimization approaches in MeNT-WiNs. Emphasis was placed on the growing significance of AI—particularly DRL—in improving resource management and service delivery within these networks. Finally, the section positioned MeNT-WiNs as a key enabler for 6G and beyond and concluded by presenting the key contributions and organizational structure of the paper.

II. FUNDAMENTALS OF AI

This section introduces the foundational principles of AI, guiding the reader through relevant concepts in AI, ML, DL, and RL, and ultimately leading to various DRL techniques. These techniques leverage the power of DL to enhance network performance and efficiency. We begin by discussing AI and its key subcategories, each adopting different learning methodologies, resulting in distinct branches.

A. AI.

AI encompasses the creation of computer systems that perform tasks that typically require human intelligence, like problem-solving, decision-making, learning, and language processing. AI strives to construct machines that can replicate human cognitive functions, allowing them to analyze data, adapt to dynamic environments, and make autonomous decisions. Techniques such as ML, DL, RL, DRL, and NLP are employed to enhance these systems, enabling them to learn from experience. Applications of AI span numerous sectors, including smart industries [83], AVs [84], [85], smart transportation [86], [87], healthcare [88], [89], [90], finance [91], [92], [93], agriculture [94], [95], [96], and smart cities [97], [98], [99], [100]. Consequently, AI is revolutionizing diverse fields, driving innovation, and reshaping the world.

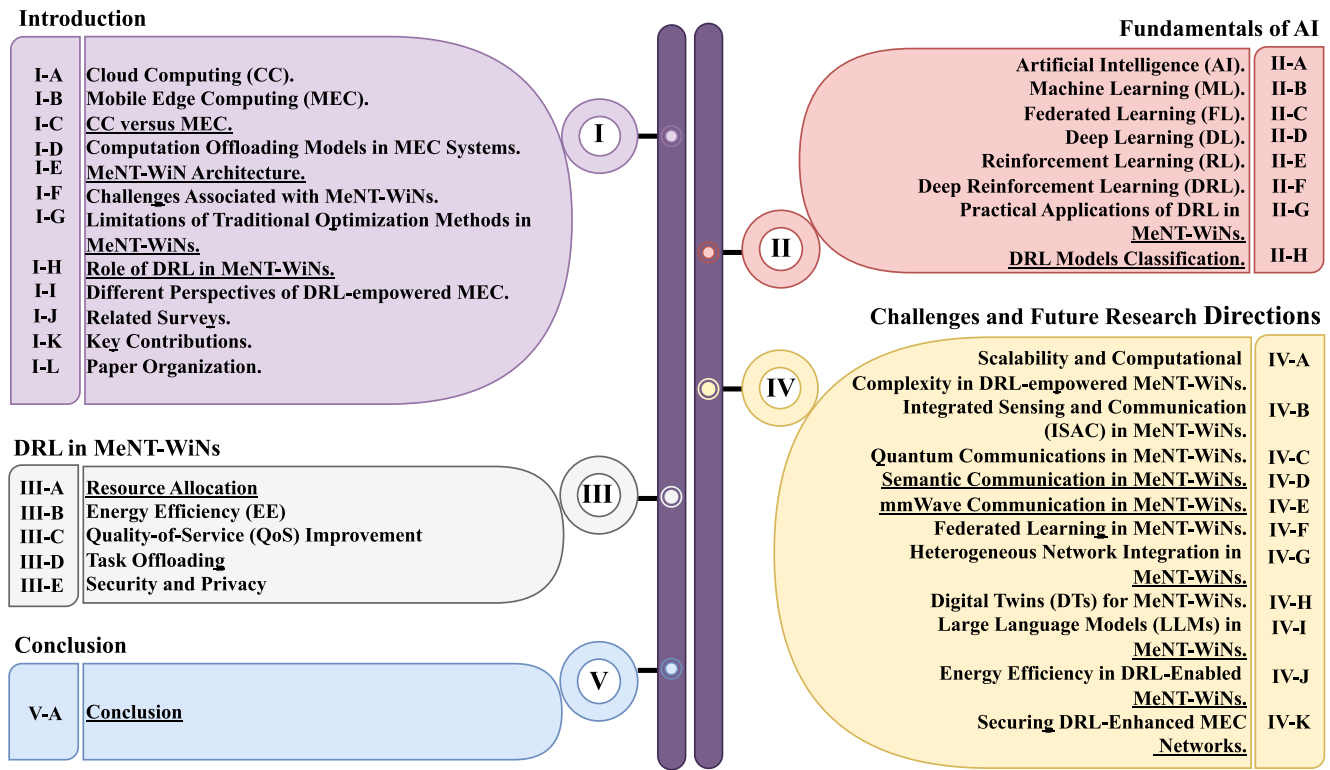


Fig. 3. Illustration of paper organization.

B. ML

As a subfield of AI, ML, aims to design algorithms that enable systems to learn from data and improve performance over time without being explicitly programmed. ML's strength lies in identifying patterns, making predictions, and extracting insights from data, thus allowing systems to adapt dynamically to new information [101], [2]. This iterative learning process makes ML particularly effective across various domains, such as healthcare, where it assists in diagnostics and personalized treatment plans [102], and finance, where it plays a key role in risk assessment and fraud detection [103]. Furthermore, ML powers image and speech recognition [104], [105], and recommendation systems, enhancing user experience across digital platforms [106]. Below, we discuss two types of ML models.

1) *Supervised Learning*: In supervised learning, a data set, containing features (or inputs) and the corresponding outputs, is supplied to the model for training, thereby called supervised learning. The model is trained over several iterations to learn how to map given inputs to outputs. The model uses the given labeled outputs to improve its performance by conducting a comparison between the actual and predicted output [107]. Supervised learning is further classified as, i) regression, and ii) classification. The regression analysis examines the relationship between a dependent variable and one or more independent variables using a statistical method. This method minimizes the error between expected and actual outputs to derive the functional mapping between inputs and outputs. Furthermore, the output label in the case of regression can assume continuous values [108]. On the other hand, in classification, the given output labels

represent distinct classes defined in image classification, etc [109].

Typically, classification problems are addressed by employing probabilistic classifiers to map output classes based on inputs, for instance, the naive Bayes classifier [110]. The supervised learning algorithms for the ML paradigm include linear regression [111], logistic regression [112], support vector machine (SVM) [113], Naive Bayes model [110], and decision tree [114]. Similarly, the supervised learning algorithms for the DL paradigm include perceptron [115], multi-layer perceptron (MLP) or fully connected NNs (FNNs) [116], extreme learning machine [117], deep residual network (DRN) [118], connected NNs [119], deconvolutional network [120], probabilistic NNs [121], recurrent NNs [122], [123], LSTM [124], reservoir computing [125], gated recurrent units, and transformers [126].

2) *Unsupervised Learning*: Unlike supervised learning, where models are trained with labeled datasets to learn a direct input-output mapping, unsupervised learning operates on datasets that contain only input features, with no associated output labels. While supervised learning aims to learn a direct mapping from inputs to known outputs, unsupervised learning seeks to discover hidden patterns, groupings, or underlying structures within the data [127]. Since the data does not come with predefined output labels, the learning objective is to identify similarities, groupings, or latent representations that describe the data distribution.

Since no labels are provided, the algorithm must analyze the input data to infer relationships or organization based on statistical properties. A key example is clustering, where the model identifies groups of similar data points based on

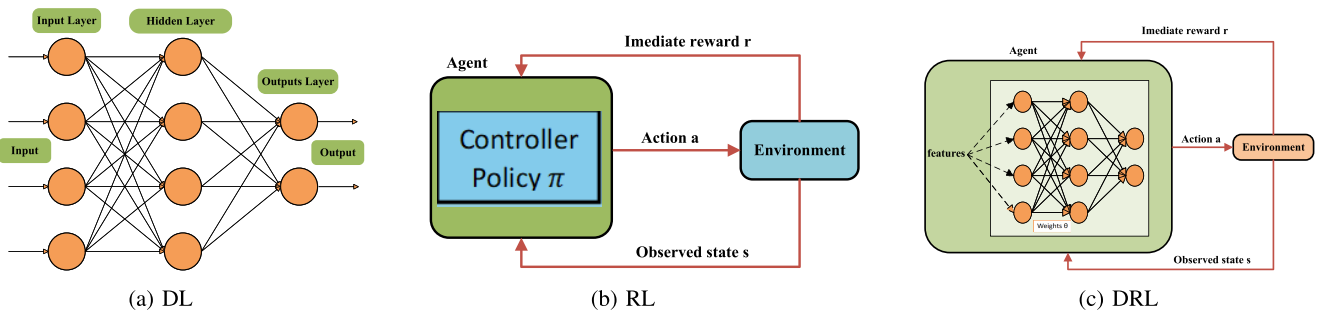


Fig. 4. Illustration of (a) DL, (b) RL, and (c) DRL.

feature similarity. Another important task is dimensionality reduction, which projects high-dimensional data into lower-dimensional representations while retaining key structural information. These unsupervised techniques are often used for data exploration, feature extraction, and as pre-processing steps in broader ML workflows.

The unsupervised learning algorithms for the ML paradigm include principal component analysis [128], K-means clustering [129], K-medoids clustering [130], Gaussian mixture model (GMM) [131], and k-Nearest neighbors (KNN) [132]. Similarly, the unsupervised learning algorithms for the ML paradigm include variational autoencoders [133], noisy autoencoders [134], sparse autoencoders [135], DBNs [136], Hopfield NN [137], Boltzmann machine [138], self-organizing maps [139], generative adversarial networks (GANs) [140], and generative diffusion models (GDMs) [141].

C. Federated Learning (FL)

In FL, the training of the model occurs across multiple devices or servers holding local data without sharing the raw data itself [142]. Clients (devices) collaborate under the supervision of a central server, which aggregates locally computed updates into a global model. FL ensures data privacy and security by keeping data localized, making it particularly effective in scenarios where privacy concerns are paramount. In edge computing environments, FL models can learn global patterns while leveraging data stored locally on each device, enabling applications such as mobile edge popularity prediction [143].

D. DL

DL, another branch of AI, builds upon ML by using NNs to address complex tasks where feature extraction is challenging. DL leverages NNs to model high-dimensional data [144], as they consist of layers of interconnected nodes, called neurons, which mimic the architecture of the human brain [145]. The multiple layers within these networks allow DL to uncover intricate patterns and relationships in data. Neurons are connected via edges assigned weights, which are iteratively optimized through backpropagation algorithms [146]. The learning process in DL often involves minimizing a cost function (representing the difference between predicted and actual outputs) using optimization techniques like stochastic gradient descent (GD) [147], [148], Newton's method [149],

and GD [150], allowing the system to learn complex representations. Fig. 4 (a) presents an illustration of DL.

E. RL

RL is an experience-driven learning approach where an agent learns by interacting with an environment rather than relying on a predefined dataset. The agent receives rewards or penalties based on the actions it takes within the environment, allowing it to improve its behavior over time [151]. The environment is often modeled using the MDP [152], which offers a mathematical framework for decision-making in stochastic scenarios. The aim of RL is to discover a policy that minimizes penalties (or maximizes cumulative rewards) over a specified time horizon, based on the agent's state. Fig. 4 (b) illustrates the basic components of an RL system.

F. DRL

DRL, first introduced in [153], merges the strengths of DL and RL, making it well-suited for complex, high-dimensional problems. While traditional RL algorithms are effective for environments with limited state and action spaces, they struggle when applied to real-world scenarios that involve continuous or high-dimensional spaces. As finding an optimal policy in such environments becomes increasingly challenging, DRL was developed to overcome this limitation by efficiently handling high-dimensional data and continuous action spaces. In DRL, the RL framework consists of a self-learning agent that maximizes long-term rewards without requiring a model of the environment. At the same time, DL, inspired by biological NNs, plays a critical role in addressing the complexities inherent in these environments, enabling the agent to extract meaningful features from large state spaces. Fig. 4 (c) illustrates the basic components of a DRL setup. DRL is particularly well-suited for tasks involving sequential decision-making [154], where the agent must make a series of decisions to optimize the expected cumulative future reward. These tasks are typically modeled as MDPs, which rely on the Markov property: the next state depends solely on the current state, and outcomes are influenced by both the agent's actions and random factors. In the following, we discuss the MDP framework, its core principles, and its role in DRL.

1) *MDP*: An MDP is a stochastic and discrete-time control process, formally defined by the tuple (S, A, T_a, R_a) , where

S represents the set of states in the environment, and A denotes the set of possible actions. Also, T_a is the state transition probability function, where $T_a(s, s') = P(s_{t+1} = s' | s_t = s, a_t = a)$ represents the probability that taking action a in state s at time t will result in a transition to state s' at time $t + 1$, while R_a is the immediate reward associated with this transition. A key assumption in MDPs is the Markov property, which states that the next state and reward depend only on the current state and action, and not on the sequence of prior states and actions. Formally, this implies that $P(s_{t+1} | s_t, a_t, s_{t-1}, a_{t-1}, \dots, s_0, a_0) = P(s_{t+1} | s_t, a_t)$. This memoryless property simplifies the modeling of sequential decision-making problems, allowing the use of dynamic programming and reinforcement learning techniques to learn optimal decision policies efficiently.

The objective of an MDP is to determine an optimal policy π^* that maps each state to an action in a manner that maximizes the expected cumulative reward over time. Policies may be deterministic (assigning a single action to each state) or stochastic (assigning a probability distribution over actions). For finite horizon MDPs, the optimization goal is given by $\max_{\pi} \mathbb{E} \{ \sum_{t=0}^T r_t(s_t, \pi(s_t)) \}$, while for the case of infinite horizon MDPs, the expected discounted total reward is given by $\max_{\pi} \mathbb{E} \{ \sum_{t=0}^{\infty} \gamma r_t(s_t, \pi(s_t)) \}$, where $\gamma \in [0, 1]$ is the discount factor that determines the relative importance of future rewards.

The MDPs can be categorized into two types: fully observable MDPs (FOMDPs) and partially observable MDPs (POMDPs) [155]. In a FOMDP, the agent has full visibility of the environment's state at each time step. However, in a POMDP, the agent only has partial information about the environment's state. A POMDP is typically represented by the tuple $(S, A, T_a, R_a, \Omega, O)$, where Ω represents the set of observations available to the agent, and O denotes the observation probabilities. The agent maintains a belief set, which is a probability distribution over the possible states. Based on its belief $b(s)$, the agent selects an action a , transitions to a new state s' , and receives a reward $r \in [R_a]$ and an observation $o \in [O]$. The agent then updates its belief about the new state s' , using the following formula [155], [156],

$$b_a^o(s') = \frac{p(o|s, a, s') \sum_s p(s'|s, a) b(s)}{\sum_{s, s'} p(o|s, a, s') p(s'|s, a) b(s)}. \quad (1)$$

Here, $p(o|s, a, s')$ is the probability that the agent observes o after taking action a and transitioning from state s to state s' , while $p(s'|s, a)$ represents the transition probability from state s to s' after action a . Similar to MDPs, the agent in a POMDP aims to find an optimal policy π^* that maximizes the expected cumulative reward over time, given by $\max_{\pi} \mathbb{E} \{ \sum_{t=0}^{\infty} \gamma r_t(s_t, \pi(s_t)) \}$. Fig. 5 depicts the illustrations of FOMDP and POMDP.

G. Practical Applications of DRL in MeNT-WiNs

Building on the foundational discussion of DRL and its capabilities, as discussed above, this section explores specific scenarios that demonstrate its practical applications in MeNT-WiNs. DRL's adaptability, real-time decision-making,

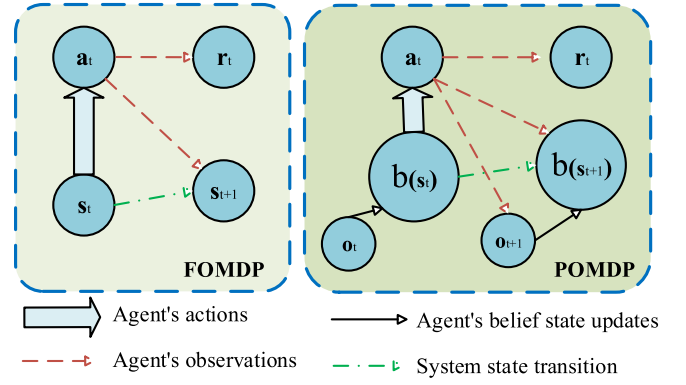


Fig. 5. Illustration of FOMDP vs POMDP.

and optimization capabilities make it particularly suited to address the complex challenges of NTN. Below, we present illustrative case studies that highlight the transformative role of DRL in optimizing the performance and resilience of MeNT-WiNs, bridging the gap between theoretical potential and real-world applications.

1) *Dynamic Spectrum Sharing in Satellite Networks*: DRL can be employed to optimize dynamic spectrum allocation in satellite constellations where spectrum availability and demand fluctuate significantly [157], [158]. For instance, consider a scenario where multiple satellites need to coordinate spectrum usage to serve ground stations across different regions with varying user densities. A DRL-based framework could dynamically allocate spectrum to satellites based on real-time demand and interference conditions, minimizing spectrum wastage and ensuring fair resource allocation. This approach outperforms static allocation methods that cannot adapt to dynamic changes in network demand.

2) *AAV Path Planning*: In a disaster recovery mission, AAVs are deployed to provide temporary connectivity and collect data from affected areas. DRL can optimize the AAVs' flight paths to maximize coverage and minimize energy consumption while dynamically routing collected data to edge servers for processing [159], [160]. By continuously adapting to environmental changes, such as wind patterns or interference, DRL ensures reliable communications and efficient use of AAV resources.

3) *EE Optimization in MeNT-WiNs*: A specific example could involve using DRL to manage power allocation across a network of LEO satellites. By monitoring the satellites' energy levels, user traffic patterns, and environmental factors such as solar panel efficiency, a DRL algorithm could dynamically adjust power distribution to balance energy consumption with communication demands [161]. This approach helps extend satellites' operational lifetime and ensures consistent QoS.

4) *Enhancing Security and Privacy in MeNT-WiNs*: In MeNT-WiNs, the dynamic and decentralized nature of satellite, HAP, and AAV networks poses significant security and privacy challenges. DRL can address these challenges by enabling adaptive anomaly detection [162], where it learns normal network behavior to identify threats like distributed denial-of-service (DDoS) attacks or unauthorized access in real time [163]. For instance, DRL can dynamically reroute

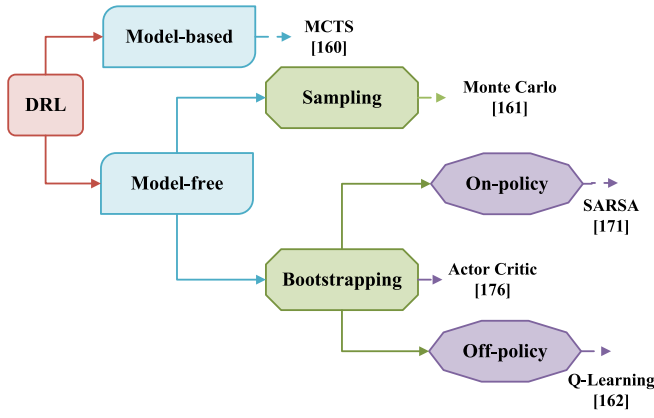


Fig. 6. Illustration of DRL methods classes.

traffic or isolate compromised nodes in a satellite network under attack [164]. Additionally, DRL enhances privacy by optimizing encryption and access control, intelligently allocating secure channels based on data sensitivity and network conditions [165].

H. DRL Models Classification

Based on the wide variety of DRL applications, several DRL algorithm variations have emerged, primarily distinguished by whether they rely on a predefined model or not. As a result, DRL methods are generally categorized into two main classes: model-based DRL methods and model-free DRL methods. Below we discuss these categories in detail. Fig. 6 depicts the different classes of DRL algorithms.

1) *Model-Based DRL Methods*: In model-based DRL methods, the agent constructs a model based on feedback obtained from the environment. This model is then used to predict the impact of actions on the states and rewards, allowing the agent to derive an optimal policy. Examples of model-based methods include AlphaZero [166], model-based RL with model-free fine-tuning (MBMF) [167], imagination-augmented agents (I2A) [168], and monte carlo tree search (MCTS) [169]. Model-based strategies typically exhibit lower sample complexity, as they reduce the need for extensive interactions with the environment. However, if the model is inaccurate, the resulting policy may be suboptimal, leading to less precise and suboptimal outcomes.

2) *Model-Free DRL Methods*: In model-free DRL methods, the agents learn optimal policies by directly interacting with the environment, without structuring a model of the dynamics of the environment. These methods are further categorized into different approaches based on how they learn from the interaction.

- *Value-Based DRL Methods*: In value-based DRL algorithms, the agent focuses on learning the value function (or state-value function), denoted as $V^\pi(s)$, rather than explicitly storing a policy $\pi(s)$. The value function estimates the expected long-term reward for each state or state-action pair. Hence, the value function $V^\pi(s)$ can be defined as a mapping from each state to its corresponding reward: $V^\pi(s) : S \Rightarrow R_a$, under policy

π , derived from the value function. Mathematically, the value function for all states is expressed as $V^\pi(s) = \sum P(\tau|\pi, s) G(\tau)$, where $P(\tau|\pi, s)$ represents the trajectory probabilities given the initial state s and policy π , and $G(\tau) = \sum_{t=0}^T \gamma r_t(s_t, \pi(s_t))$. Accordingly, the optimal state-value function is defined as $V^*(s) = \max_\pi V^\pi(s), \forall s \in S$.

Another key function, known as the Q-function (or action-value function) and denoted as $Q^\pi(s, a)$, maps state-action pairs to long-term rewards: $Q : S \times A \Rightarrow R_a$. The key distinction between the action-value function and the state-value function is that the Q-function incorporates the initial action as part of its input, while the state-value function does not. The mathematical formulation of the action-value function for all states and actions is given by $Q^\pi(s, a) = \sum_\pi P(\tau|\pi, s, a) G(\tau)$, where $P(\tau|\pi, s, a)$ represents the trajectory probabilities given the initial state s , action a , and policy π . The optimal action-value function is thus defined as $Q^*(s, a) = \max_\pi Q^\pi(s, a), \forall s \in S, \forall a \in A$.

In value-based DRL, the agent's objective is to learn an optimal policy that maximizes the expected cumulative reward, which is expressed as

$$\pi(s) = \max_\pi V^\pi(s), \forall s \in S. \quad (2)$$

Similarly, given an optimal Q-value function, the corresponding optimal policy is expressed as

$$\pi = \max_{a \in A} Q^*(s, a), \forall s \in S. \quad (3)$$

Thus, in value-based DRL, the task of finding an optimal policy translates into approximating the optimal action-value function, which can typically be accomplished using either the temporal difference (TD) method or MC estimation.

The MC estimation method periodically updates the action-value function and refines the policy based on the principle of generalized policy iteration [170]. The MC method improves the policy by generating a new greedy policy that selects the action with the highest long-term reward for each state $s \in S$, as given by

$$\pi_{k+1}(s) \leftarrow \max_{a \in A} Q^{\pi_k}(s, a). \quad (4)$$

On the other hand, the TD approach periodically updates the action-value function by incorporating concepts from dynamic programming (DP), allowing the agent to learn from incomplete episodes through bootstrapping. Specifically, the action-value function is updated using the immediate reward along with the estimated value of the next state. TD learning leverages the Markov property, where the next state s_{t+1} is determined solely by the current state s_t , enabling iterative updates after every decision epoch. This allows for online learning without waiting for episodes to conclude.

In contrast, MC estimation relies on returns from complete episodes, providing lower bias but requiring significant experience for convergence. It also suffers from high variance and is suitable mainly for episodic

TABLE V
SUMMARY OF DRL ALGORITHMS

Algo.	Key Characteristics	Applications
Value-Based DRL Algorithms.		
QL	A TD learning algorithm designed for discrete state and action spaces, which iteratively updates action-value estimates to converge towards the optimal policy.	Applied in robotics, industrial automation, gaming, and finance.
DQL	Utilizes deep NNs to approximate the Q-function, enabling effective learning in high-dimensional state spaces.	Appropriate for MDPs that involve discrete state and action spaces.
PDQL	Prioritizes experiences with larger TD errors, assigning them higher priority for replay, which improves sample efficiency and accelerates learning.	Appropriate for MDPs prioritizing experiences.
DDQL	Reduces the overestimation of Q-values by employing two separate networks to independently estimate the target and current Q-values, improving stability and overall performance.	Used in gaming, robotics, and finance, with effective DQ-value estimation for decision-making.
ADQL	A parallelized version of DQL, where multiple agents interact with separate instances of the environment, enabling faster learning through asynchronous updates to the Q-network.	Effective for managing asynchronous updates in distributed environments.
Dis-DQL	Distributes the training process across multiple machines, allowing for faster training and improved scalability to larger environments and datasets.	Excels at capturing uncertainty and enhancing performance in stochastic environments.
Due-DQL	Decomposes the Q-value into two components: an advantage function and a state value function, leading to improved learning efficiency and more accurate Q-value approximations.	Effective for MDPs with large action spaces.
DQLNN	Introduces noisy layers into the NN to encourage exploration, improving learning efficiency by dynamically adjusting exploration strategies.	Appropriate for MDPs with large action, and state spaces.
Rainbow	Combines several advanced techniques, such as PER, DQL, dueling networks, and noisy layers, to achieve state-of-the-art performance in DQL.	Ideal for complex problems with high-dimensional state and action spaces and uncertainty.
SARSA	An on-policy TD learning algorithm that updates action values based on the outcomes of its own actions, incorporating the next action taken and the associated reward.	Used in discrete state-action spaces for on-policy learning and optimal policy convergence.
Policy-Based DRL Algorithms.		
REINFORCE	Utilizes DNNs to train the policy function, optimizing policy performance through direct estimation of gradients based on the rewards received.	Appropriate for problems with discrete action spaces.
TRPO	Applies constraints on policy updates to ensure stability and prevent large, destabilizing changes, thereby improving the reliability of policy optimization.	Effective for large-scale continuous control problems.
PPO	Utilizes a clipped surrogate objective function to constrain policy updates, preventing large, disruptive changes and promoting stable and reliable policy improvement.	Effective for high-dimensional or continuous action spaces, like robotics and games.
Actor-Critic DRL Algorithms.		
A2C	Integrates policy gradient and value-based methods by simultaneously learning action selection and value estimation, which enhances both efficiency and stability in training.	Appropriate for problems with continuous action spaces.
A3C	Combines policy gradient and value-based methods with multiple parallel agents, improving sample efficiency and training stability through asynchronous updates.	Ideal for problems needing asynchronous training for efficient exploration.
SAC	Enhances stochastic policies and value function learning by incorporating entropy regularization, which promotes exploration and robustness in continuous action spaces.	Appropriate for problems requiring stable learning and robust exploration.
DDPG	Tailored for continuous action spaces, utilizing deterministic policy gradients and experience replay to achieve stable and efficient learning.	Used for continuous action spaces needing stable learning and efficient exploration.
CER-DDPG	Extends DDPG by ensuring the inclusion of the most recent experience tuples in batch selection, thereby enhancing sample efficiency and learning stability.	Used where sample efficiency and fast convergence are crucial.
PER-DDPG	Integrates PER with DDPG, giving higher priority to experiences with larger TD errors during replay.	Beneficial for problems requiring improved sample efficiency and faster convergence.
MADDPG	Extends DDPG for multi-agent settings, enabling centralized training and decentralized execution while supporting cooperation in complex environments.	Appropriate multi-agent problems in complex environments.
TD3	Uses twin critics and delayed policy updates to stabilize training and improve policy robustness.	Ideal for continuous action spaces where stability and robustness in learning are key for efficient exploration and decision-making.
RDPG	Uses recurrent NNs in the policy network to learn deterministic policies in time-sensitive or sequential observation environments.	Effective for sequential decision-making with crucial temporal dependencies in time-sensitive scenarios.
D4PG	Enhances DDPG by incorporating distributional RL techniques and parallelized training across multiple actors, facilitating efficient exploration in large continuous action spaces.	Used in continuous action spaces where stability and robustness in learning are crucial for efficient exploration and decision-making.

(or discrete) MDPs. TD, which predicts the next state, exhibits higher bias but lower variance, making it more robust and suitable for continuous MDPs. Contemporary DRL algorithms favor TD learning due to its adaptability and superior learning capabilities. Some value-based DRL algorithms include Q-learning (QL) [171], deep QL (DQL) [153], Asynchronous DQL (ADQL) [172], distributional DQL (Dis-DQL) [173], double DQL

(DDQL) [174], prioritized DQL (PDQL) [175], dueling DQL (Due-DQL) [176], DQL with noisy nets (DQLNN) [177], rainbow [178], and state-action-reward-state-action (SARSA) [179]. A summary of these algorithms is provided in Table V.

- *Policy-Based DRL Methods:* In policy-based methods, agents do not utilize a distinct value function. Instead, they directly optimize a policy. By employing policy

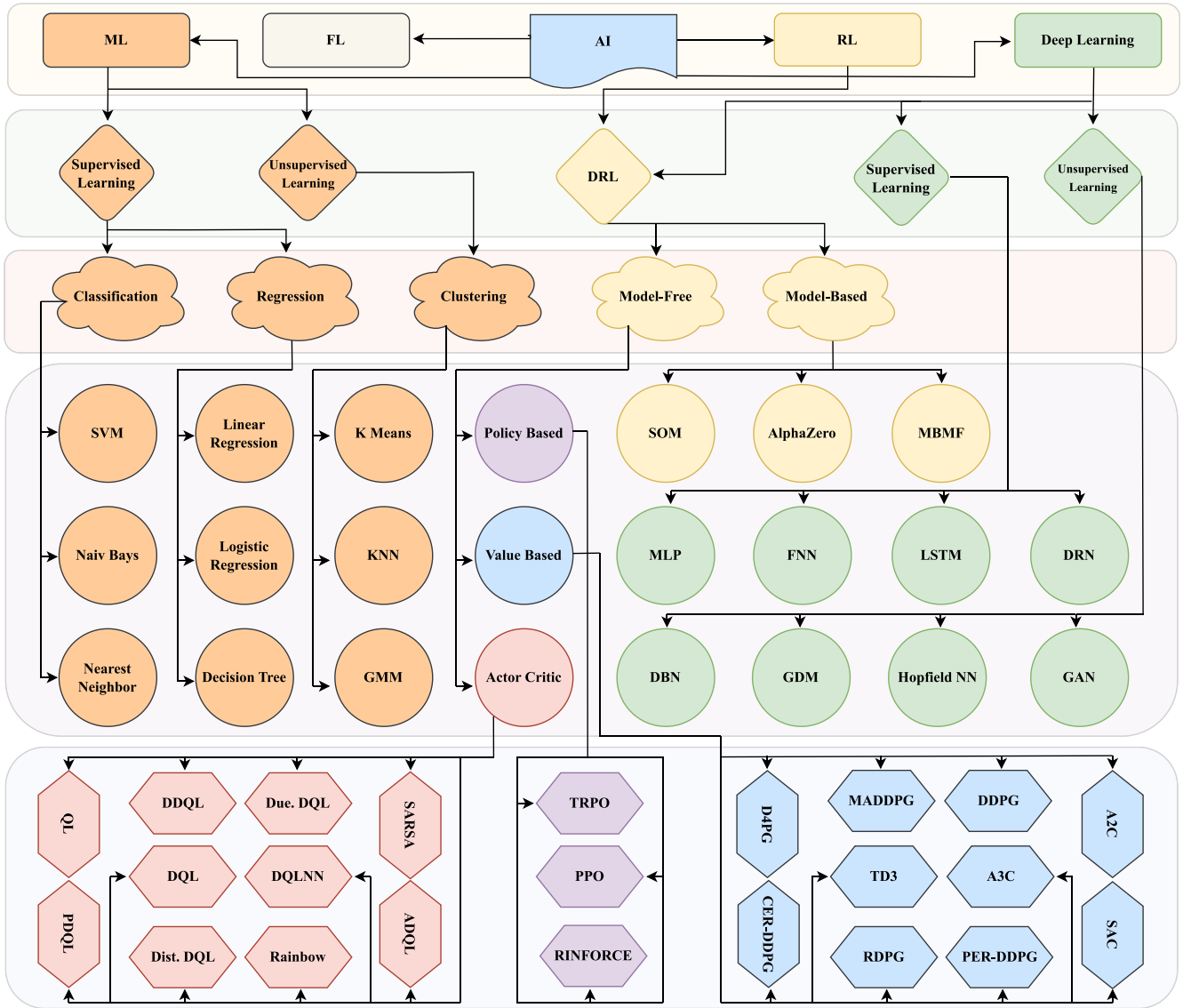


Fig. 7. AI Taxonomy.

gradient techniques, the agent begins with an initial policy function and iteratively refines it over multiple episodes, selecting actions without relying on a value function. Policy-based methods are particularly suited for optimization problems involving continuous action spaces, where the number of possible action values is finite. There are two main types of policies: (i) stochastic policies, which map a state to a probability distribution over the action space, denoted as $P(a|s) = \pi_\theta(a|s)$, and (ii) deterministic policies, which map each state directly to an action, represented as $a = \pi_\theta(a|s)$. The objective function for a policy-based method is expressed as $J(\theta) = \sum_\tau P(\tau|\pi_\theta) G(\tau)$, where $P(\tau|\pi_\theta)$ denotes the probability of trajectories under the policy π_θ . Thus, the core problem in a policy-based algorithm is to find the optimal parameter θ that maximizes $J(\theta)$.

For differentiable parameters, the objective function can be optimized by taking its derivative with respect to θ to maximize the long-term reward. This is given by [180],

$$\nabla_\theta J(\theta) = P_{\pi_\theta} \left[G(\tau|s, a) \nabla_\theta \ln \pi_\theta(a|s) \right], \quad (5)$$

where $P_{\pi_\theta} \cdot$ represents the distribution of trajectories under policy π_θ , and $G(\tau|s, a)$ is the long-term reward for a trajectory τ given state s and action a .

Policy-based DRL methods utilize gradient-based optimization, and the objective function is maximized through gradient ascent. The policy parameters are updated at each step according to

$$\theta_{t+1} = \theta_t + \alpha \gamma^t \left[G(\tau|s_t, a_t) \nabla_\theta \ln \pi_\theta(a_t|s_t) \right], \quad (6)$$

where α is the learning rate. Examples of policy-based DRL methods include REINFORCE [181], proximal policy optimization (PPO) [182], and trust region policy optimization (TRPO) [183]. A summary of these policy-based DRL algorithms is provided in Table V, and a taxonomy tree is provided in Fig. 7.

- *Actor-Critic Methods:* In this class of DRL algorithms, the attributes of both policy and value-based approaches

are combined. These methods employ two NNs: the actor network and the critic network. The actor network operates similarly to the NN in policy-based methods, while the critic network functions like those in value-based methods. In the actor-critic architecture, the actor network is responsible for selecting actions that maximize the expected cumulative reward over time. Whereas, the critic network evaluates the quality of the actions taken by the actor by computing the value function. A feedback loop is integral to these methods, where the critic provides feedback to the actor by assessing the expected cumulative reward of being in a particular state and following the actor's policy thereafter. In essence, the critic network optimizes the parameters of the value functions based on the actor's policy. Simultaneously, the actor network updates its policy parameters using the value function evaluations provided by the critic. Examples of actor-critic algorithms include advantage actor-critic (A2C) [184], soft actor-critic (SAC) [185], asynchronous A2C (A3C) [172], deep deterministic policy gradient (DDPG) [186], combined experience replay DDPG (CER-DDPG) [187], prioritized experience replay DDPG (PER-DDPG) [188], multi-agent DDPG (MADDPG) [189], twin delayed DDPG (TD3) [190], recurrent DPG (RDPG) [191], and distributed distributional DDPG (D4PG) [192]. A summary of these actor-critic algorithms is provided in Table V.

In summary, this section provided a comprehensive overview of key AI paradigms, beginning with foundational concepts in AI and ML, and progressing through DL, FL, and RL. A particular emphasis was placed on DRL, highlighting its ability to enable autonomous and adaptive decision-making in dynamic wireless environments. We also surveyed practical applications of DRL within MeNT-WiNs, demonstrating its relevance in addressing critical tasks such as resource management, task offloading, and energy optimization. Finally, the section presented a classification of DRL models commonly employed in MeNT-WiNs, providing insight into their design characteristics and suitability for various network scenarios.

III. DRL IN MENT-WiNs

MEC is envisioned as a promising solution to address the high demand for high-throughput and low-latency applications in wireless communication networks. In the context of non-terrestrial environments such as AAVs, satellite networks, and aerial drones, MEC offers the potential to bring storage and computation capabilities closer to the edge of the network, enabling faster response times and improved user experiences. However, the effective utilization of MEC resources in NT-WiNs faces challenges due to factors like intermittent connectivity, limited bandwidth, and dynamic environmental conditions. In the recent past, DRL has gained traction as a powerful tool to address certain challenges faced by wireless communication networks, for instance, resource allocation optimization, enhancing EE, improving QoS, task offloading, addressing security concerns, and network optimization. Together with MEC, DRL can offer the opportunity to enhance

the performance of wireless communication networks, specifically the NTN.

This section provides a detailed review of existing literature on DRL-assisted MEC in NT-WiNs particularly, related to resource allocation, EE, QoS improvement, task offloading, and security and privacy. This section explains how DRL-enabled MEC can bring transformative changes to communication systems in non-terrestrial settings, unlocking new opportunities for applications and services by synthesizing insights from diverse studies.

A. Resource Allocation

In general, resource allocation is a key issue in MeNT-WiNs, where computing, storage, and communication resources have to be used efficiently to optimize network performance for diversified applications and services. Recently, DRL has been considered an attractive paradigm for dynamic resource allocation in the network edge, as it has the capability to learn and discover the optimal policies that eventually lead to optimal decision-making while interacting with the environment. This subsection highlights a review of the existing literature related to DRL-assisted resource allocation strategies in MeNT-WiNs. We look into how DRL algorithms are being implemented for effective resource allocation adaptable to dynamic network conditions, traffic patterns, and user demands with the view to maximizing resource utilization, latency minimization, and optimizing network efficiency. The following subsection synthesizes insights from studies conducted within a wide range, with the aim of presenting an extended review to understand the state-of-the-art approaches in DRL-enhanced resource allocation for MeNT-WiNs.

The authors in [193] formulated a joint optimization problem that addressed both computation offloading and resource allocation in a MEC-empowered integrated aerial-terrestrial vehicular network. They modeled the problem as an MDP, which captured the stochastic nature of the environment, including the variability in user demands, communication channel conditions, and computational workloads. Using their concept of minimizing the total overhead of computation in the network, they proposed a DDQL-based solution. In the system, every vehicle corresponded to some intelligent agent that interacted with an unknown and dynamic environment to learn its best policy in an adaptive manner. The DDQL-based solution harnessed DQL to optimize primary parameters, such as offloading computation decisions and resource allocation strategies. One of the strengths of this approach is adaptability; it can effectively handle changes within the system, adapting to dynamic fluctuations in traffic patterns and resource availability as they arise in the network. In [194], the authors presented a DRL-driven architecture for a mixed-integer non-linear programming (MINLP) problem to jointly optimize resource allocation and computation offloading in an aerial-to-ground (A2G) network. The proposed approach is model-free DRL, which enabled the system to learn the optimal policy in a dynamic environment by using past experience rather than relying on knowledge of the underlying system model. In the context of this work, the AAV worked as an aerial edge

computing node that offloaded some computation tasks from edge IoT devices to ease up their computational burden. The goal was to find the minimum of both computation delay and energy consumption through dynamic offloading and resource allocation. The results demonstrated that the proposed scheme has much superior efficiency compared to existing methods in terms of latency reduction, enhanced energy saving, and enhanced scalability of the system.

A multi-agent RL (MARL) algorithm was proposed in [195] by the authors to optimize radio resources, computational capacities, and AAV trajectories for hierarchical NTN specifically designed for disaster relief services. The main aim of the proposed framework is to maximize the number of satisfactory connections with the constraint of tight end-to-end latency constraints. The authors used an LSTM-based mechanism LSTM that allowed the system to encode and exploit the prior observation as well as the prior action in the transaction history of the network for better accuracy and efficiency in decision-making. The experimental results, particularly regarding connection quality and system scalability, demonstrated that the proposed algorithm significantly outperformed its competitors, especially in scenarios with substantial increases in AAVs and IIoT nodes. Such scalability in the solution makes it appropriate for large-scale deployments in disaster scenarios where speed and flexibility are critical for rapid response. The work in [196] presented a satellite-assisted ground-integrated network architecture that integrated LEO satellites, AAVs, and cloud servers to provide edge computing services for ground users when there is no infrastructure. The authors formulated several optimization problems that minimized both energy consumption and latency within the network. They proposed a MADRL algorithm, to address the very challenging problem of hybrid discrete-continuous action spaces and focused on long-term optimization objectives. In detailed simulation studies, the proposed algorithm has demonstrated superior performance for each agent compared to some benchmark methods. The achieved simulation results were found to show superior improvements in terms of energy consumption, latency, and the execution time of the task. The proposed solution proved effective in a satellite-assisted edge-computing environment, addressing the specific needs of such systems.

In [197], the authors proposed a multi-task resource scheduling (MRS) framework, based on a DRL foundation that was devised for minimizing the total energy consumption of an enhanced AAV-assisted MEC system with reconfigurable intelligent surfaces (RIS). RISs are utilized to improve the quality of communications between UEs and AAVs and to more efficiently use resources. It used an A2C structure that covered a multi-head agent, a multi-task controller, and a light wolf search algorithm to optimize different parameters such as the positioning of AAVs, phase-shifted diagonal matrices, computation offloading, and resource allocation. The results of the study showed that the multi-agent RS (MARS) framework outperformed benchmarks significantly. In [198], a GEO satellite-assisted vehicle-to-vehicle (V2V) communication network was discussed as an optimization problem for jointly optimizing task assignment, scheduling,

and resource management for communication and computing tasks by minimizing the long-term weighted sum latency of multitasking operations. To address this issue, the authors proposed a DRL-based solution, which modeled the problem of task scheduling and assignment as an MDP and then used a DQL algorithm to solve it. The simulations conducted verified both the convergence and effectiveness of the algorithm in terms of latency reduction and overall improvement in system performance. This solution has great potential to be deployed in V2V communication networks where low-latency and high-reliability communication are of paramount importance for the safe and efficient running of vehicles.

The work in [199] introduced a new task offloading and resource allocation strategy in the context of space-terrestrial integrated optical transmission (STIOT) networks. The authors combined DDPG and Lagrange multiplier algorithms to attain optimal resource allocation and used the Karush-Kuhn-Tucker (KKT) conditions. The network's objective was defined in terms of minimizing latency and energy consumption. Leveraging powerful optimization techniques, the authors proposed a novel approach that significantly reduced task execution times and energy consumption, making it highly suitable for space-terrestrial networks requiring real-time processing and resource efficiency. In [200], the authors proposed an end-to-end video analytics framework that joined MEC and blockchain technology to optimize transaction throughput and reduce the latency of applications based on Internet-of-AVs (IoAVs), like surveillance video. The proposed solution used an A3C-based algorithm that efficiently handled offloading video and resource allocation within a decentralized network. Indeed, with the A3C-based algorithm, an optimum solution for the most suitable reward function was discovered because it considered important factors such as the size of data, CPU cycles, and latency, which allowed maximum potential throughput for system efficiency with a minimal delay in real-time video surveillance in AV networks.

B. EE

EE and sustainability continue to be the major issues in the design and operation of MEC systems deployed in NTNs. DRL presents an exciting route for optimizing energy consumption in the MEC-enabled non-terrestrial environment by learning adaptive policies that minimize energy use while satisfying the prescribed performance. In this subsection, we review the literature on DRL-based solutions for EE enhancement in MEC deployments under diverse non-terrestrial scenarios: satellite networks, aerial drones, and AAVs. We will also review how dynamically DRL algorithms operate on power consumption management, optimize resource utilization, and perform task scheduling to improve EE without degrading any network performance. This section will review the state-of-the-art techniques in leveraging DRL for energy-efficient MeNT-WiNs through synthesizing insights across a range of studies.

In [161], the authors introduced a novel architecture for integrated satellite-aerial-terrestrial relay networks (ISATRNs) that utilized HAPs with AAVs equipped with RIS to enhance

EE and optimize system performance. The authors aimed to maximize the system's ergodic rate by jointly optimizing AAV trajectory, RIS phase shift, and active transmit beamforming matrix while considering AAV energy consumption constraints. To address this inherently complex optimization problem, they proposed a DRL-based framework using an improved LSTM-DDQL algorithm. The results indicated that their proposed method outperformed traditional DDQL algorithms in terms of single-step exploration average reward values and other performance metrics, demonstrating significant potential for practical applications in enhancing the reliability and efficiency of next-generation ISATRNs. The authors in [201] presented an optimization framework for enhancing EE in multiple aerial RIS-assisted downlink communication systems. By leveraging the advantages of AAV deployment and RIS technology, the study addressed poor line-of-sight issues between UE and BS. The authors proposed a joint optimization problem encompassing RIS deployment, reflecting elements' on/off states, phase shift, and power control, which is complex due to its mixed-integer, non-convex, and NP-hard nature. This problem is decomposed into three sub-problems, solved alternately using successive convex approximation (SCA), actor-critic-PPO, and the whale optimization algorithm. Extensive simulations demonstrated the proposed algorithms' efficacy, significantly improving EE compared to several benchmark schemes. Ying Chen et al. [202] proposed a distributed online task offloading and resource allocation algorithm to minimize the energy consumption of ground devices by reducing the drift-plus-penalty function in each time slot in an aerial-MEC system supported by HAPs and AAVs, using stochastic optimization techniques and game theory. The distributed online task offloading and resource allocation algorithm is shown to have the lowest energy consumption and smallest queue backlog compared to other algorithms, reducing energy consumption by 15.73%–19.09% and queue backlog by 15.99%–88.84%.

In [203], the authors proposed a multi-AAV-assisted MEC network incorporating a digital twin (DT) layer to minimize the weighted energy consumption of mobile users and AAVs, while ensuring that the task delay requirements are met. The study employed a MARL approach based on multi-agent PPO (MAPPO) with a Beta distribution and an attention mechanism to achieve optimal computation offloading policies. The performance of the MAPPO algorithm was evaluated against other RL baselines, demonstrating its effectiveness in reducing energy consumption and improving both fairness and network stability. The work in [204] introduced a AAV-aided MEC architecture for vehicular networks, which utilized line-of-sight massive multiple-input-multiple-output (mMIMO) transmission to minimize the weighted total energy consumption of vehicles and aerial roadside units (ARSUs). The energy consumption for vehicles and ARSUs in each time slot, including offloading, downloading, and computation tasks, was formulated as a multi-variable optimization problem aimed at minimizing total energy consumption, subject to physical layer constraints, time slot scheduling, and task allocation. The study found that massive MIMO can mitigate

distance-dependent path loss and reduce unnecessary ARSU mobility.

Ei et al. [205] proposed a two-stage, multi-AAV-assisted edge computing system, where AAVs provided both computing and relaying services to mobile devices. They formulated a joint task offloading, communication, and computation resource allocation problem to minimize the energy consumed by AAVs and mobile devices. In the proposed system, mobile devices offloaded tasks to AAVs, which then relayed part of the tasks to terrestrial BSs for further processing, while minimizing energy consumption under resource constraints and task deadlines. Simulations demonstrated that the approach reduced network energy consumption and outperformed baseline methods. In [206], the authors presented a novel priority-based DRL approach, aimed at optimizing the trajectory planning, communication scheduling, charging schedules, and task offloading (PD-TCCT) of a AAV in a MEC network to maximize the EE of AAV. Simulations demonstrated the efficacy of the AAV-MEC network, showcasing improved EE and a reduced energy consumption ratio compared to various baseline approaches.

C. QoS Improvement

Ensuring high-quality service delivery is paramount in NTN, where factors such as latency, data rate, and reliability significantly impact user experience and application performance. MEC combined with DRL presents a promising approach to enhance QoS in such environments. In this subsection, we explore the literature on leveraging DRL techniques to improve various aspects of QoS in MeNT-WiNs. By analyzing a range of studies, we aim to provide insights into how DRL-based strategies can effectively optimize QoS metrics, leading to enhanced network performance and user satisfaction.

In [207], the authors proposed a comprehensive optimization framework designed to address both latency and EE in a MEC-empowered satellite-based IoT (SAT-IoT). The network consisted of multiple satellites, gateways, and a multitude of IoT devices, all working in tandem to support a wide range of communication and computation tasks. The core objective of this framework was to jointly optimize key parameters, including offloading decisions, user association, computing resource allocation, communication resource management, and the allocation of minimal energy at the UE level. By combining these aspects, the authors aimed to solve the overarching problem of efficient resource utilization and task execution in SAT-IoT environments. This complex optimization problem was formulated as a dynamic mixed-integer problem, which could be computationally intractable in its full form. To address this, the problem was divided into two sub-problems, simplifying the solution process without sacrificing overall performance. The authors implemented their proposed joint user association and offloading decision with an optimal resource allocation (JUAOD-ORA) algorithm and conducted extensive simulations to evaluate its performance. The results demonstrated that the algorithm significantly outperformed existing baseline methods across various

metrics, including total cost, latency, and energy consumption. These findings highlight the JUAOD-ORA algorithm's ability to efficiently balance the trade-offs between computational power and energy resources, making it a highly effective solution for SAT-IoT systems.

In another study, Pervez et al. [208] developed an optimization approach aimed at reducing both energy consumption and latency in a multi-AAV-assisted MEC network. In this network configuration, several AAVs equipped with MEC servers collaborated with a ground BS to provide computational services to mobile users. The AAVs facilitated offloading of tasks from mobile users, thus easing the computational burden on local devices. The authors' approach to optimization is grounded in the block descent method, which breaks the overall problem into smaller tasks that are tackled sequentially. These tasks include but are not limited to, offloading decisions, MEC server selection, transmission power allocation, AAV path planning, and CPU frequency allocation for task processing. To solve these challenges, the authors proposed a game-theoretic framework for task offloading, ensuring that users' tasks are efficiently distributed across available resources. For power distribution, the geometric water-filling method was employed, which optimizes the allocation of transmission power based on channel conditions. Furthermore, SCA techniques were utilized to address the AAV trajectory planning problem, ensuring that AAVs follow the most efficient paths while maintaining connectivity with ground devices. Lastly, GD methods were applied to optimize the CPU frequency allocation, allowing for dynamic adjustments based on computational demand. The integration of these diverse optimization techniques resulted in a highly adaptable and efficient system capable of minimizing both energy consumption and latency.

In another study, Liu et al. [209] explored the use of multi-AAV-enabled MEC systems to enhance maritime communication. Given the challenging nature of maritime environments, where connectivity is often limited or unstable, the focus of this research was on developing AAV-centric strategies that optimize key aspects of the network's performance. The study specifically addressed issues related to trajectory design, resource allocation, and computation offloading. The primary goal of these strategies was to minimize latency and reduce energy consumption while maintaining high levels of service quality. To achieve this, the authors proposed a DDPG algorithm to optimize both the trajectory design and the configuration of virtual machines (VMs) for a Tactical-AAV. This problem was formulated as a non-convex mixed-integer nonlinear optimization problem, making it challenging to solve directly. However, the DDPG algorithm demonstrated its effectiveness by outperforming a DQL-based algorithm in reducing latency. This performance improvement is attributed to DDPG's ability to process continuous input data and generate continuous action outputs, making it more suitable for real-time AAV trajectory control. In [210], the authors introduced a federated DQL (F-DQL) algorithm designed to minimize system latency in a federated blockchain-empowered vehicular network (FBVN). In this network, AVs serve as the key components, training DNNs

locally on their onboard systems. To enhance overall network efficiency, AVs periodically share their learned parameters with edge servers, thereby reducing system-wide latency and simultaneously maintaining data privacy. This decentralized approach ensured that raw data never leaves the local devices, mitigating the risk of data breaches. The AVs functioned as intelligent agents, continuously sensing the state of the wireless channels, the quality of the datasets they generated, and the current packet error rates. Based on this information, AVs made real-time decisions regarding offloading strategies, wireless channel selection, and the adjustment of CPU-cycle frequency. The performance of the F-DQL algorithm was evaluated against three variants: F-DQL without offloading selection, F-DQL without wireless channel selection, and F-DQL without CPU-cycle frequency selection. Comparative analysis showed that the complete F-DQL algorithm achieved significant reductions in system latency, clearly outperforming all three variants.

The study in [211] explored a DQL approach aimed at optimizing latency in MEC-augmented LEO satellites. By integrating MEC servers into LEO satellites, the system enabled users to offload computation tasks, thereby improving the overall performance of the IoT network. The proposed optimization method combines the Lagrange multiplier technique with a DQL, enabling the system to make optimal task-offloading decisions. The objective was to minimize total system latency while ensuring the smooth operation of IoT applications. Simulation results indicated that the proposed combined approach resulted in a notable enhancement in IoT network performance, reducing the latency associated with task execution and data processing. Another relevant study, [212], focused on latency optimization for hybrid GEO-LEO satellite-assisted IoT networks. The authors proposed a task offloading and multidimensional resource allocation (TOMRA) algorithm, which integrates DRL with convex optimization techniques to minimize both task offloading and processing latency. The system architecture included both inter-satellite cooperation (between GEO and LEO satellites) and satellite-ground cooperation, creating a complex but highly efficient network capable of meeting the demands of IoT applications. Simulation results demonstrated that the proposed TOMRA algorithm consistently outperformed benchmark schemes, making it a promising solution for latency-sensitive IoT applications in hybrid satellite networks.

D. Task Offloading

Task offloading, the process of migrating computation-intensive tasks from mobile devices to edge servers or cloud resources, plays a crucial role in optimizing resource utilization and enhancing performance in NTN. MEC combined with DRL offers a promising approach to dynamically manage task offloading decisions based on network conditions, user requirements, and application characteristics. In this subsection, we explore the literature on DRL-driven task offloading strategies in MeNT-WiNs. By examining various offloading policies, algorithms, and optimization techniques, we aim to gain insights into how DRL can effectively optimize task

offloading decisions to improve network efficiency, reduce latency, and enhance user experience in MEC-empowered non-terrestrial environments.

The authors introduced a multi-agent energy-efficient joint trajectory and computation offloading scheme in [213], specifically designed for AAV-enhanced MEC networks. This proposed scheme integrates a sophisticated optimization-embedding multi-agent DRL algorithm that dynamically adapts to changing user demands in the network. The key objective is to achieve both EE and service fairness across the network by addressing the complex interplay between the AAVs' flight paths and the users' offloading requirements. To achieve this, the approach optimized fairness-based EE for AAVs, particularly in scenarios involving intensive computation offloading tasks. In such scenarios, AAVs not only operate individually but also collaboratively devise strategies for trajectory control, admission control, and computation resource allocation. The ultimate goal is to maximize the long-term fairness-based EE, ensuring that the overall network resources are used optimally over time. The authors proposed an optimization-embedding multi-agent due. DQL (OMADDQL) algorithm, which, through simulations, demonstrated superior performance compared to other benchmark algorithms, particularly with respect to fairness-based EE. Additionally, the algorithm's performance was influenced by parameters such as the learning rate, which had a significant impact on the convergence behavior of the average accumulated reward. This makes the OMADDQL a highly adaptive and effective solution for managing computation offloading in AAV-assisted MEC systems, especially in scenarios with rapidly changing dynamics. In [214], the authors proposed a novel cooperative data sensing and computation offloading scheme specifically designed for AAV-assisted mobile crowdsensing systems. This scheme focuses on jointly optimizing multiple critical factors, including flight direction, flight distance, the proportion of tasks to offload, and the selection of offload servers. By considering all of these parameters together, the authors aimed to maximize the overall system utility—which in this context is a measure of how effectively the system uses its resources to complete tasks. The method employed an actor-critic-based DRL framework, which leveraged a multi-agent DRL algorithm designed to minimize control-response delay and enhance overall system performance. In comprehensive performance evaluations, the proposed scheme outperformed existing baseline algorithms, demonstrating significant improvements in terms of system reward, task completion rate, and average cost per task. Moreover, it was observed that the proposed method achieved faster convergence rates and improved trajectory planning, showcasing its ability to adapt to the demands of a real-time AAV-assisted mobile crowdsensing system.

Furthermore, the authors in [215], presented a novel approach for data offloading and computation services within the context of intelligent transportation systems (ITS). The proposed approach leverages satellite access networks, which use LEO satellites and CubeSats as independent MEC servers. The authors proposed a cooperative MAPPO algorithm, enhanced with an attention mechanism to optimize key

decision-making processes such as offloading task selection, computing resource allocation, and bandwidth management. The main objective of this scheme is to minimize both the mean service time and the overall service cost. Simulation results indicated that the proposed algorithm significantly outperformed baseline methods, particularly in terms of reducing latency and minimizing the cost of service, thus demonstrating its applicability to modern ITS applications where low-latency communication is critical. In [216], the focus shifted towards a hierarchical AAV swarm-enabled MEC system, where the authors introduced a DRL-based computation offloading scheme. This scheme was designed to minimize the weighted sum of task execution delay and energy consumption, considering both the characteristics of the tasks being offloaded and the underlying network dynamics. The study revealed that, in comparison to conventional offloading methods, the proposed scheme achieved faster algorithm convergence, reduced offloading costs, and significantly improved performance in terms of both task execution delay and energy consumption. These findings underscore the effectiveness of DRL-based solutions in managing the complexities of hierarchical AAV systems, where multiple UAVs work together to optimize the offloading and execution of tasks across a distributed network.

The study in [217], introduced a distributed DRL approach aimed at optimizing task offloading and service migration in satellite edge computing (SEC) networks. Given the unique challenges posed by SEC environments—such as limited service coverage and resource constraints—the authors focused on minimizing the long-term average delay for all users within the network. The solution involved a double due. DQN (D3QN)-based distributed scheme, which was carefully designed to balance the demands of task offloading while managing service migration effectively. The study compared the proposed scheme's performance with other methods such as round-robin scheduling, traditional CC, and various DRL-based benchmarks [218]. Results indicated that the D3QN-based approach consistently outperformed these methods in scenarios involving complex resource management and offloading tasks.

In another relevant study, [219], the authors examined a computation offloading problem within a HAP-aided vehicular network. In situations where ground BS are either overloaded or inaccessible, such as in rural or disaster-stricken areas, the authors proposed a DRL-based intelligent offloading strategy. The proposed strategy employed the DDPG algorithm to minimize both the total energy consumption and task execution latency for all vehicles in the network. By optimizing offloading decisions in real-time, the algorithm allows vehicles to efficiently offload computation tasks to HAPs, reducing the overall network load and improving service delivery times. Finally, in [220], the authors introduced a MEC framework that integrated both AAVs and a macro BS to deliver computation offloading services to wireless users. The study formulated an optimization problem aimed at minimizing the total system cost, which included key factors such as task execution delay, energy consumption, and bandwidth utilization costs. The proposed AAV-assisted computation offloading scheme for

MEC, built on the DQL algorithm, was developed to optimize the offloading policies in real-time. Performance evaluations under a variety of conditions—such as varying arrival rates of renewable energy and the number of work units covered by each AAV—demonstrated that the proposed scheme achieved near-optimal performance in terms of minimizing total system costs, making it a highly efficient solution for dynamic, AAV-assisted MEC networks.

E. Security and Privacy

Security and privacy are assumed to be the topmost concerns in MeNT-WiNs, where transmitting sensitive data and communication operations are prone to various kinds of threats and attacks. This section presents one of the promising frameworks that integrate MEC with DRL in view of improving security and privacy. In general, with the use of DRL algorithms, MEC systems can adaptively respond to evolving security threats and effectively mitigate privacy risks while guaranteeing efficiency in network operation. This section reviews the literature on DRL-based solutions for ensuring security and privacy in MeNT-WiNs. Specifically, it examines existing studies on how DRL techniques detect or prevent security breaches, safeguard user privacy, and ensure data transmission integrity and confidentiality in non-terrestrial environments.

The authors in [221] have equipped a multi-agent DRL-based framework for all disjointed spectrum access, energy control, and resource management allocation to vehicles in vehicular edge computing (VEC) networks while aiming to achieve an optimal system processing delay and at the same time maintaining the security and reliability of V2V communication links. A DDQL is proposed to optimize the joint SORA scheme in a VEC network, where multiple vehicle users concurrently learn how to minimize system processing delay and proceed to secrecy probability maximization. On the other hand, Consul et al. [222], devised an efficient secure MEC system using AAVs and non-orthogonal multiple access (NOMA) schemes and further adopted an FL algorithm to bolster the security of the system through blockchain technology. This study concentrated on the physical layer of security, while blockchain technology is recommended to be useful for ensuring the transparency and traceability of the data supplied by the user as a consensus mechanism.

The authors in [223] explored a privacy-enhanced blockchain computing offloading framework for satellite-ground DT networks that integrated multi-agent reinforced learning and Lyapunov stability theory to reduce execution energy and privacy time overhead. A distributed RL algorithm is employed to minimize the privacy protection costs and the power consumption of the system whereby a Lyapunov drift plus penalty is utilized on nested FL to update the weights of actor networks to compute optimal offloading strategies. Their proposed approach outperformed the benchmark strategies by achieving the least execution energy overhead, the highest quantity of bits processed, and the least privacy time overhead. Lu et al. [224], investigated secure transmission methods for multi-UAV-assisted MEC networks by employing single-agent and MARL schemes. They aimed to determine

optimal AAVs' locations and accomplish overall system load offloading without compromising the overall system utility by combining several users' tasks which have some varying requirements on computing hardware's leftover energy and delay. The QL-based single agent task offloading scheme incorporates a Q table used to store each action taken, the state of that action, and the expected reward of that action being performed for each state-action pair. The agent picks such action that corresponds to a maximum value of Q. The multi-agent scheme based on Nash QL is employed to optimize the secure offloading, where each agent updates its Q-table according to the feedback of the environment through continuous interaction with the environment. The multi-agent method outperformed the single-agent and random offloading schemes in terms of system and energy utility, however, it accumulated a longer total delay.

F. Network Optimization

Network optimization is essential for maximizing the efficiency, performance, and reliability of communication systems in non-terrestrial environments, where dynamic conditions and resource constraints pose significant challenges. MEC combined with DRL is a promising paradigm toward network operation optimization for overall improvements in system performance. MEC systems can leverage DRL algorithms to enable dynamic and adaptive reconfiguration of network settings, resource allocation, and routing strategies, ensuring that the network can adapt to changing demands and environmental conditions over time. In this regard, we review the related literature about network optimization strategies driven by DRL in the case of MeNT-WiNs. We present how several DRL techniques can be leveraged to optimize network efficiency, scalability, and robustness in non-terrestrial environments, surveyed from existing studies, paving the way for advanced communication systems and services.

In [225], the authors proposed a MEC-empowered air-ground integrated network (AGIN) framework where AAVs act as edge servers to provide efficient computation services for IoT devices and UE. A MADDPG-based strategy was developed to jointly optimize task deployment and offloading, addressing factors such as task placement, replacement, offloading decisions, resource allocation, and power control. The MADDPG framework achieved convergence and demonstrated superior performance in simulations compared to the baseline algorithms. In [226], the authors introduced a DRL-based algorithm aimed at maximizing the sum computation rate in a wireless-powered multi-access point edge computing network consisting of multiple HAPs and IoT devices. These IoT devices can harvest radio frequency energy from HAPs for local data processing or offload tasks to the HAP edge servers. The proposed algorithm optimized the offloading decision and time allocation in this wireless-powered edge computing system. Simulation results indicated that the algorithm performed comparably to supervised learning techniques without needing optimal samples and outperformed algorithms such as DQL and AC in terms of computation rate and computational complexity, especially in large-scale network settings.

The authors in [227], explored a DRL-based online resource management framework for a multi-user, multi-server MEC system assisted by AAVs, with the goal of minimizing the system's average power consumption while meeting long-term constraints on queue stability and computational delay. The authors formulated a stochastic optimization problem for managing radio and computation resources, transforming it into a more tractable form, which they solved using a Lyapunov optimization framework and a DRL-based online optimization algorithm. The proposed DRL-based resource management (DRLRM) method is compared with benchmark schemes in two scenarios, demonstrating its effectiveness in reducing power consumption and stabilizing queues, and its convergence performance is evaluated under different settings of the hyperparameter k and queue length threshold. In [228], the authors investigated the MEC paradigm for AI applications in air-ground-integrated wireless networks. They proposed a AAV-empowered MEC system that optimizes DNN model decisions, communication resource allocation, and AAV trajectory control to minimize service latency while ensuring learning accuracy and energy consumption. An iterative approach is used to optimize a AAV-empowered MEC system's model decision, resource allocation, and trajectory control simultaneously. The Iterative method alternates between solving three subproblems, each of which is convex or approximate convex. The proposed scheme achieved lower service latency and better accuracy performance compared to benchmark schemes, and simulation findings show that it can offer great service performance while adhering to energy consumption, latency, and accuracy requirements.

Liu et al. [229], explored a learning-based resource management optimization algorithm for AAV-assisted MEC against dynamic jamming attacks, where a jammer aims to disrupt the communication link between the AAV and a BS by emitting interference power. The AAV and jammer are modeled as agents in an MDP to jointly optimize AAV hover point selection, task collection time ratio, transmission power, channel selection, and task offloading ratio to improve communication and computing efficiency while considering total AAV power and channel availability constraints. The proposed scheme demonstrated a high exploration rate to handle diverse environments while preventing overfitting, and achieved optimal efficiency with improvements in reward and task throughput by over 15% and 30%, respectively, compared to existing schemes. In [197], the authors explored a MARS framework based on DRL to minimize energy consumption in a dynamic MEC system aided by AAVs and RISs. The MARS framework, which combined an actor-critic structure with a multi-head agent, a multi-task controller, and a light wolf search algorithm, optimized the positions of AAVs, phase-shifted diagonal matrix, computation offloading, and resource allocation. The superior performance of the proposed algorithm is observed through various simulations when compared with various existing benchmark schemes.

Qian et al. [230] proposed a multi-domain, joint trajectory optimization and resource allocation, for a AAV-empowered maritime IoT network. They aimed to optimize the total

energy consumption of unmanned surface vehicles (USVs) and AAVs by optimizing AAV trajectory, computation, and communication resources. The authors investigated a dual-layer framework to solve the optimization problem by transforming the original problem into a convex optimization problem and then solving it using a dual-primal algorithm, and subsequently solving the first-layer problem using a DRL algorithm. In order to resolve the first formulation of the optimization problem, a double-layered segmented decomposition framework was used. The suggested Top-Algorithm provided a sub-optimal solution of 88% in terms of optimality and hence reduced the computational burden such that it outperformed other existing DRL algorithms and baseline schemes in the total energy consumption of the AAV-empowered M-IoT network. The authors in [231] investigated a semi-distributed multi-agent FRL (MAFRL) algorithm to optimize power control, user association, and resource allocation, in a AAV-empowered MEC system. The proposed algorithm aimed to optimize the resource allocation and offloading decision in a AAV-empowered MEC system, where each UE learns its own policy to optimize the overall power consumption while satisfying the latency constraint. The MAFRL algorithm achieved close performance to a centralized MARL algorithm while protecting user privacy with Gaussian differentials and reducing operation time by 23%. Table VI summarizes the literature surveyed for exploring the applications of DRL in MeNT-WiNs.

In summary, this section explored the application of DRL to address key functional requirements in MeNT-WiNs. It examined how DRL enables intelligent and adaptive resource allocation, efficiently managing spectrum, power, and computation resources in highly dynamic MEC-enhanced non-terrestrial environments. The section further discussed strategies to enhance EE by leveraging DRL for energy-aware decision-making and sustainable network operation. For QoS improvement, DRL was shown to effectively balance latency, reliability, and throughput under varying network conditions. In the context of task offloading, DRL-based methods were presented as viable solutions for optimizing computation distribution across edge and non-terrestrial nodes. Additionally, the section addressed security and privacy concerns, highlighting how DRL can support threat detection, secure access control, and privacy-preserving mechanisms in distributed MeNT-WiNs. Collectively, these discussions demonstrate the versatility and promise of DRL in enabling robust, efficient, and intelligent MeNT-WiNs.

IV. CHALLENGES AND FUTURE RESEARCH DIRECTIONS

While there have been significant advances in leveraging DRL for MEC in NTN, MeNT-WiN environments present novel and complex challenges. The unique characteristics of NTNs—such as long propagation delays, dynamic topologies, and limited onboard resources—demand specialized approaches that transcend those designed for terrestrial networks. In this section, we explore the primary challenges and outline future research directions for DRL-assisted MeNT-WiN systems. We

TABLE VI
SUMMARY OF WORKS ON DRL-AIDED MeNT-WiNs

Cat.	Ref.	Algo.	Key contributions
Resource Allocation	[193], 2022	DDQL	Optimized resource allocation and computation offloading in MEC-empowered integrated aerial-terrestrial vehicular network.
	[194], 2021	DDPG	Minimized computation delay and energy consumption in a multi-UAV-assisted massive IoT network via dynamic resource allocation and computation offloading.
	[195], 2024	MARDPG	Maximized satisfactory connections and ensures end-to-end latency in NTN for disaster relief services through optimized radio resources, computing capacities, and UAV trajectories.
	[196], 2024	SAC	Reduced energy consumption and latency in a satellite-assisted ground-integrated network by solving complex optimization problems with a multi-agent DRL algorithm.
	[197], 2023	MARS	Minimized energy consumption in a UAV-empowered MEC-assisted system with RISs by optimizing UAV positions, resource allocation, computation offloading, and phase-shifted diagonal matrix.
	[198], 2021	DQL	Minimized long-term weighted-sum latency in GEO satellite-enhanced V2V communication by optimizing scheduling, resource management, and task assignment, jointly.
	[199], 2023	DDPG	Introduced a novel resource allocation strategy in STIoT to minimize latency and energy consumption using DDPG and Lagrange multiplier algorithm with KKT conditions.
	[200], 2022	A3C	Integrated MEC and blockchain technologies to optimize transaction throughput and reduce latency in IoAV for video surveillance applications using an A3C-based algorithm for video offloading and resource allocation.
EE	[164], 2024	DDQL	Proposed a novel architecture for ISATRNs using HAPs with UAVs equipped with RIS to enhance EE and optimize system performance.
	[201], 2024	AC-PPO	Enhanced EE in multiple aerial RIS-assisted downlink communication systems, addressing poor line-of-sight issues between UE and BS by leveraging UAV and RIS.
	[202], 2023	Game Theory	Minimized the energy consumption of ground devices in an aerial-MEC system supported by HAPs and UAVs, utilizing stochastic optimization and game theory.
	[203], 2023	MAPPO	Minimized the weighted energy consumption of UAVs and mobile users, while guaranteeing task delay requirements, using MAPPO with Beta distribution in a multi-UAV-empowered MEC network.
	[204], 2021	CO	Presented a novel UAV-aided MEC architecture for VNs utilizing line-of-sight mMIMO transmission to minimize the weighted total energy consumption of vehicles and ARSUs.
	[205], 2022	BSUM	Minimized power consumption of MDs and UAVs by jointly optimizing task offloading, computation and communication, resource allocation in a multi-UAV-empowered dual-stage MEC network.
	[206], 2022	PD-TCCT	Optimized communication scheduling, trajectory planning, task offloading, and charging schedules of a UAV in a MEC network to maximize the EE of the UAV.
	[207], 2020	DQL	Optimized latency and energy in MEC-empowered SAT-IoT networks via jointly optimizing offloading decision, user association, computing, and communication resource allocation.
QoS Improvement	[208], 2024	SAC-GD	Minimized latency and power consumption in a multi-UAV-empowered MEC network by jointly optimizing MEC server selection, UAV trajectory, transmission power, task offloading, and CPU frequency allocation.
	[209], 2022	DDPG	Employed multi-UAV-enabled MEC systems to improve maritime communication by optimizing computation offloading, trajectory design, and resource allocation to minimize power consumption and latency.
	[210], 2022	F-DQL	Minimized system latency in an FBVN by leveraging AVs as intelligent agents for wireless channel selection, offloading strategy, and onboard CPU-cycle frequency selection.
	[211], 2023	DQL	Explored latency optimization in MEC-augmented LEO satellites, integrating MEC servers into LEO satellites to enable task offloading and enhance IoT network performance.

(Continued)

present key challenges such as scalability, computational complexity, adaptability to dynamic space environments, security vulnerabilities, and the integration of emerging technologies. Moreover, we highlight the potential of DRL in enabling novel communication services and enhancing network resilience for MeNT-WiNs.

A. Computational Complexity and Scalability in DRL-Empowered MeNT-WiNs

Applying DRL in MeNT-WiNs presents several formidable challenges stemming from the inherent characteristics of non-terrestrial environments. The spatiotemporal variability [232] introduced by channel fading, mobility, and interference leads

TABLE VI
Continued. SUMMARY OF WORKS ON DRL-AIDED MeNT-WiNs

Cat.	Ref.	Algo.	Key contributions
Task Offloading	[212], 2023	PPO	Investigated latency optimization for hybrid GEO-LEO satellite-empowered IoT networks through a TOMRA combining DRL and convex optimization.
	[213], 2024	MADDQL	Optimized fairness-based EE for UAVs in computation offloading by collaboratively devising computation resource allocation, admission control, and trajectory control strategies.
	[214], 2022	MADDPG	Maximized overall system utility in UAV-empowered mobile crowdsensing by jointly optimizing flight distance, flight direction, server offload selection, and task offloading proportion.
	[215], 2024	MAPPO	Optimized bandwidth and computing resource allocation with offloading task selection, in ITS data offloading and computation services using LEO and CubeSats.
	[216], 2023	DDQL	Minimized the energy consumption and the weighted sum cost of task execution delay in a hierarchical UAV swarm-enhanced MEC system.
	[217], 2024	D3QN	Minimized the long-term average delay of all users in a SEC network through a D3QN-based distributed task offloading optimization scheme.
	[219], 2022	DDPG	Minimized total power consumption and task execution latency in a HAP-empowered VN through a DRL-based intelligent offloading strategy using the DDPG algorithm.
Security and Privacy	[220], 2020	DQL	Minimized total system cost, including power consumption, delay, and bandwidth cost, in a UAV-empowered MEC framework using a DRL-driven computation offloading policy.
	[221], 2023	DDQL	Minimized system processing delay and maximizes secrecy probability in VEC networks by optimizing joint power control, spectrum access, and computation resource allocation using a multi-agent DRL-driven scheme.
	[222], 2024	FL	Enhanced the security of a UAV-assisted MEC system using NOMA schemes and blockchain technology for transparency and decentralization, focusing on physical layer security.
	[223], 2023	MAFL	Minimized execution energy consumption and privacy time overhead in satellite-ground DT networks using a Lyapunov-based multi-agent federated RL approach.
Network Optimization	[224], 2023	QL	Optimized UAV deployment and secure offloading to maximize system utility using single-agent and MARL methods.
	[225], 2024	MADDPG	Jointly optimized task deployment and offloading processes in a MEC-empowered air-ground integrated network using UAVs as edge servers.
	[226], 2024	DQL	Maximized the sum computation rate in a wireless-powered MEC by optimizing offloading decisions and time allocation for IoT devices.
	[227], 2023	DRLRM	Minimized average power consumption in a multi-user, multi-server MEC system assisted by a UAV through a DRL-based online resource management framework that ensures queue stability and computational delay constraints.
	[228], 2023	DNN	Minimized service latency in a UAV-empowered MEC system by optimizing communication resource allocation, DNN model decision, and UAV trajectory control, jointly.
	[229], 2024	D3QN	Enhanced communication and computing efficiency in UAV-assisted MEC under dynamic jamming attacks by jointly optimizing UAV hover point selection, task collection time ratio, transmission power, channel selection, and task offloading ratio.
	[202], 2023	DRL-MARS	Minimized energy consumption in a dynamic MEC system with UAVs and RISs by optimizing phase-shifted diagonal matrix, UAV positions, resource allocation, and computation offloading.
	[230], 2023	DDPG	Minimized total energy consumption of USVs and UAVs in M-IoT networks by jointly optimizing UAV trajectory, computation and communication resource allocation.
Security and Privacy	[231], 2021	FL	Introduced semi-distributed MAFRL algorithm for optimizing user association, resource allocation, and power control in UAV-empowered MEC systems.

to a highly dynamic and enlarged state space that agents must learn and adapt to in near real-time. Furthermore, URLLC requirements impose strict constraints on decision latency and reliability, necessitating highly sample-efficient learning and minimal exploration. The large-scale deployment of heterogeneous agents—such as AAVs, IoT nodes, and satellites—with diverse roles and capabilities results in massive scalability issues and a combinatorial explosion in the joint action space [233]. Additionally, the limited onboard

computational and energy resources of these agents restrict the frequency of model updates and necessitate lightweight inference architectures. Moreover, partial observability [234] and delayed or noisy feedback, due to decentralized coordination and asynchronous communications, further complicate policy learning, making conventional DRL approaches less effective in these settings. Below we discuss computational complexity and scalability challenges in DRL-assisted MeNT-WiNs.

1) *Computational Complexity*: Applying DRL in non-terrestrial networks, particularly in MeNT-WiNs, introduces substantial computational complexity due to the high-dimensional state and action spaces. This complexity is further amplified by the dynamic topology, mobility of AAVs and satellites, and time-varying wireless channel conditions. Such characteristics impose significant computational demands during both the training and inference phases of DRL algorithms [235], [236]. The primary source of computational load arises from the DNN operations—specifically, the forward and backward propagations across multiple layers. To illustrate this, consider the DDPG algorithm, where the actor network is composed of an input layer, several hidden layers, and an output layer.

Suppose that S denotes the dimensionality of the state space, I is the number of neurons in the input of each layer, H is the number of hidden layers, O is the number of neurons in the output of each hidden layer, and A is the dimensionality of the action space, respectively. The computational complexity for processing the input layer is given by $S \times I$, for the hidden layers by $H \times I \times O$, and for the output layer by $O \times A$. Thus, the overall complexity of the actor network during inference is expressed as [237], [238]

$$C_{\text{DDPG}} = S \times I + H \times I \times O + O \times A + A.$$

To analyze the scalability of DRL algorithms, we express this using Big O notation. In computational complexity theory, Big O notation $\mathcal{O}(\cdot)$ characterizes the upper bound of an algorithm's runtime or space usage with respect to input size. It allows us to identify which operations dominate the overall computational cost as the network scales. For DRL models, this is especially useful in understanding how architectural depth, layer width, and the dimensions of state and action spaces impact performance. Given that $H \times I \times O$ generally dominates the other terms, i.e., $H \times I \times O \gg S \times I, O \times A + A$, for deep networks, the above complexity expression simplifies to

$$\begin{aligned} C_{\text{DDPG}} &= \mathcal{O}(S \times I + H \times I \times O + O \times A + A) \\ &\approx \mathcal{O}(H \cdot I \cdot O). \end{aligned}$$

This indicates that the computation is primarily driven by the number of hidden layers and their interconnections.

This complexity analysis highlights how the neural architecture and the size of the state-action space significantly influence the inference cost. In MeNT-WiNs, where real-time responsiveness and resource limitations are critical—particularly for AAVs and LEO satellites—such computational demands can be prohibitive. Therefore, strategies such as model compression, knowledge distillation, and edge-assisted DRL are essential to enable efficient and scalable deployment under constrained environments.

2) *Scalability*: Regarding scalability, one of the primary challenges is the state-space explosion problem in DRL models [239]. As the number of devices and operational states in MeNT-WiNs grows, the dimensionality of the state and action spaces increases exponentially. This leads to higher computational requirements and prolonged convergence times

for DRL algorithms. The continuously changing network environment necessitates frequent model updates, compounding the computational complexity. Additionally, the computational overhead of training and inference in DRL models, particularly those employing DNNs, can overwhelm resource-constrained platforms like satellites and AAVs. This often results in processing delays and diminished responsiveness. Communication latency between distributed MEC nodes, exacerbated by long distances and high-latency links inherent to NTN, further hampers real-time decision-making.

Energy consumption presents another significant challenge. The intensive computational demands of DRL algorithms can rapidly deplete the limited power resources of AAVs and satellites, necessitating efficient power management strategies to ensure prolonged operation and sustained performance. DRL training can be highly energy-intensive and computationally demanding, especially when large deep neural networks are involved [235]. While offline training in data centers mitigates on-board computational costs, real-time adaptation through online or continual learning—often necessary in dynamic network environments—reintroduces significant overhead. AAVs and satellites operating in MeNT-WiNs must frequently update policies in response to changing channel conditions, mobility events, and handovers, which may necessitate partial on-board learning or frequent synchronization with ground servers [240], [241]. Even during deployment, deep model inference can be computationally expensive, particularly for policy optimization methods (e.g., PPO, SAC) and value-based approaches (e.g., DQN), which rely on iterative updates, gradient computations, and experience replay buffers. In resource-constrained environments, even a forward pass through a deep neural network can rapidly drain the limited power resources of AAVs and satellites, particularly if inference is performed at high frequency [242], [243].

Furthermore, DRL systems frequently encounter issues related to data sparsity and imbalance, as heterogeneous nodes in MeNT-WiNs may provide non-uniform or limited data, impairing model training and decision-making capabilities. Moreover, the highly dynamic topologies of AAVs and satellites result in continuously changing communication links and coverage areas, requiring frequent policy adaptation in real time [244]. Furthermore, intermittent connectivity due to atmospheric conditions and orbital mechanics disrupts the learning feedback loop, making consistent policy updates challenging [241]. Unlike terrestrial networks, where DRL agents can rely on stable infrastructure, MeNT-WiNs require asynchronous learning and predictive modeling to mitigate these disruptions. Another critical challenge is real-time adaptation, as rapid and unpredictable wireless channel variations necessitate near-instantaneous decision-making. Additionally, traditional DRL methods, which involve complex iterative computations, may introduce unacceptable latency. Thus, lightweight policy networks, edge-assisted DRL, or hybrid model-based/model-free approaches are required to ensure computational feasibility and responsiveness.

A promising direction for further research could be employing advanced techniques is essential. Model compression methods [245] can help reduce the computational burden of

DNNs, enhancing their feasibility for resource-constrained platforms. FL [142] enables collaborative DRL model training across distributed nodes without extensive data exchange, preserving bandwidth and ensuring data privacy. Distributed DRL frameworks [217] decentralize decision-making, allowing for efficient resource allocation in large-scale, dynamic environments. By leveraging these advanced strategies, it becomes possible to address scalability and computational complexity bottlenecks, enhancing the efficiency, adaptability, and resilience of MeNT-WiNs. These solutions pave the way for more robust and intelligent operations, enabling MeNT-WiNs to meet the demands of next-generation communication networks.

B. Integrated Sensing and Communication (ISAC) in MeNT-WiNs

ISAC is a new paradigm in wireless communication networks where the same infrastructure serves for both communication and environmental sensing. Clearly, such duality is helpful in the context of MeNT-WiNs, where the spectrum, power, and computation resources are scarce and must be exploited efficiently. However, sensing-communication integration brings about significant challenges, especially in NTN where platforms like satellites and AAVs operate in dynamic environments with resources that fluctuate. The central issue to be addressed is how to make the various trade-offs between sensing accuracy and communication performance in general, especially when both functions need to be executed in parallel.

One of the most promising avenues for future research involves the application of DRL algorithms to ISAC-enhanced MeNT-WiNs, where DRL can be instrumental in optimizing the dual functionalities of sensing and communication, enabling intelligent decision-making to balance competing demands while ensuring fairness. As an example, a certain DRL algorithm may decide to give more emphasis to communication tasks during low sensing demand, while in other critical scenarios, such as natural disasters and search-and-rescue missions, the resources should be shifted toward sensing functions to grab essential data about the environment.

C. Quantum Communications in MeNT-WiNs

Quantum communication holds immense promise for transforming MeNT-WiNs by leveraging the unique principles of quantum mechanics [246]. While its primary focus has been on enhancing data security through Quantum key distribution [247], its applications extend far beyond secure communication. For instance, quantum communication can enable precise synchronization among NTN components [248], such as satellites, HAPs, and AAVs, facilitating better coordination in dynamic environments. Additionally, quantum-enhanced sensing offers opportunities to improve the accuracy and sensitivity of measurements [249], benefiting applications such as environmental monitoring and disaster management. Furthermore, distributed quantum computing [249], [250] powered by quantum entanglement can enable collaborative problem-solving across NTN nodes,

addressing computationally intensive tasks in remote or infrastructure-limited regions.

Despite these opportunities, the integration of quantum communications into MeNT-WiNs presents significant challenges. The mobility and dynamic nature of NTN components, such as AAVs and HAPs, complicate the stability and scalability of quantum networks. DRL can be leveraged to optimize dynamic resource allocation and ensure energy-efficient quantum protocols tailored for MeNT-WiN environments, which are also critical to overcoming the limitations of power-limited platforms and ensuring reliable NTN operations. DRL-based strategies can optimize resource allocation, including bandwidth, power, and computational resources, to ensure the efficient operation of quantum communication systems under severe constraints. Addressing these research directions should unlock the full potential of quantum communication in MeNT-WiNs, paving the way for intelligent, scalable, and energy-efficient quantum-enabled MeNT-WiNs.

D. Semantic Communication in MeNT-WiNs

In non-terrestrial environments, where bandwidth is often constrained, and network conditions are highly variable, traditional methods of communication that focus on the accurate transmission of bitstreams may become inefficient. Semantic communication, which shifts the focus from transmitting raw data to conveying the underlying meaning or semantics of the information, becomes a viable solution to meet the scarcity of resources in NTNs. Specifically, in the context of MeNT-WiNs, this approach can greatly reduce transmission overhead and improve system efficiency, as only the most essential information is sent across the network, minimizing the use of limited resources [251]. Additionally, by prioritizing the transmission of the meaning (semantics) of data rather than raw data, semantic communication aligns seamlessly with the vision of intelligence as a service in 6G networks.

The integration of semantic communication into MeNT-WiNs presents three primary challenges: security, resource allocation, and multimodality [252]. In MeNT-WiNs, DRL can significantly enhance semantic communication by addressing these key challenges. It optimizes bandwidth and power usage, ensuring efficient resource utilization in constrained environments, and prioritizes data modalities based on their semantic importance and network conditions. Additionally, it strengthens security by dynamically adapting to evolving threats and assists the encoding and decoding processes by learning effective semantic representations [253]. These capabilities make DRL an essential enabler for realizing the full potential of semantic communication in MeNT-WiNs.

E. Millimeter-Wave (mmWave) Communication in MeNT-WiNs

mmWave technology is one of the key enablers for high-throughput communication in MeNT-WiNs, especially in AAV-empowered platforms where line-of-sight transmission is predominant. However, deploying mmWave in MeNT-WiNs is not without its challenges, as it is burdened with rapidly changing channel conditions, mobility of satellites and AAVs,

and interference due to environmental factors. The high-frequency nature of mmWave signals enables much faster data transmission compared to common wireless communication signals but makes it easier to attenuate and block the mmWave signal compared to lower-frequency signals.

DRL could be a strong driving force in overcoming such challenges by dynamically learning and adapting to find optimal strategies for mmWave transmission. DRL can be employed to optimize beamforming [254], interference management [255], and resource allocation [256] in real-time to guarantee such network performance where the conditions fluctuate unpredictably. For instance, future research should develop DRL models to optimize routing by minimizing interference and efficiently managing power to reliably deploy mmWave communications in any space-based or aerial communication system.

F. FL in MeNT-WiNs

FL introduces a decentralized approach to ML, where multiple nodes collaborate to train models without sharing their raw data. This is especially beneficial in MeNT-WiNs, where platforms such as satellites, AAVs, and HAPs generate vast amounts of data across wide geographical areas. In MeNT-WiNs, FL enables distributed intelligence, allowing these platforms to learn from diverse datasets while preserving privacy and reducing the need for centralized data processing [257].

Nevertheless, applying FL in MeNT-WiNs presents unique challenges. The heterogeneous nature of MeNT-WiNs, combined with varying levels of connectivity, latency, and processing power, complicates the synchronization of learning processes across distributed nodes. Communication overhead is also a significant concern, as frequent updates to global models can strain bandwidth and energy resources. Future research should focus on developing adaptive FL algorithms tailored to the intermittent and resource-constrained nature of MeNT-WiNs, optimizing both communication efficiency and model synchronization. Hierarchical FL architectures [258] could also be explored, where learning processes are coordinated across different layers of the network, improving scalability and reducing communication overhead.

G. Heterogeneous Network Integration in MeNT-WiNs

The integration of heterogeneous networks, including satellites, AAVs, and terrestrial systems, is a critical challenge in the development of MeNT-WiNs. This integration aims to enable seamless wide-area communication across platforms with distinct operational characteristics and constraints. The complexity arises from managing handovers [259], resource optimization, and ensuring interoperability among these diverse systems, necessitating advanced and intelligent algorithms capable of real-time decision-making.

Future research should leverage DRL to address these integration challenges. DRL can enable the development of intelligent frameworks that optimize resource allocation in real-time, handle dynamic handover processes across heterogeneous platforms, and ensure robust interoperability.

Specifically, DRL models should focus on tailoring solutions to the unique characteristics of satellites, AAVs, and terrestrial systems, enabling energy-efficient, low-latency, and high-throughput communications.

H. DTs for MeNT-WiNs

DT technology holds immense potential in transforming the management and optimization of MeNT-WiNs. By creating virtual representations of network elements, DTs enable real-time integration of data from non-terrestrial platforms such as satellites, HAPs, and AAVs. These DT models facilitate advanced simulations, proactive monitoring, and data-driven decision-making [260], [261], all crucial for addressing the complexities inherent in the dynamic and heterogeneous environments of MeNT-WiNs. For example, DT-enabled MeNT-WiNs can dynamically optimize resource allocation, improve EE through adaptive power management, and serve as platforms for testing new technologies before deployment, minimizing risks and accelerating innovation.

Nevertheless, the implementation of DTs in MeNT-WiNs present significant challenges, including handling the massive volume of real-time data from diverse sources, ensuring seamless two-way information exchange, and addressing the computational overhead of creating and maintaining high-fidelity virtual models. Additionally, the integration of DTs with MeNT-WiNs require lightweight, scalable frameworks capable of real-time operation. DRL can play a transformative role in addressing these challenges and enhancing the functionality of DTs within MeNT-WiNs. DRL can be employed to develop intelligent decision-making frameworks for real-time optimization of resource allocation and traffic management. Furthermore, DRL algorithms can facilitate adaptive model compression, reducing the computational burden of DTs while preserving fidelity. DRL-driven solutions can also optimize the interaction between physical network entities and their virtual counterparts, enabling robust fault detection [262], predictive maintenance [263], and enhanced energy management. Future research should focus on creating DRL-assisted frameworks that seamlessly integrate DTs into MeNT-WiNs, ensuring efficient and resilient network operations.

I. Large Language Models (LLMs) in MeNT-WiNs

Integrating LLMs into MeNT-WiNs represents a promising research direction with the potential to enhance network autonomy, intelligent management, and efficient communication across heterogeneous platforms. LLMs excel at processing and analyzing vast streams of unstructured data, enabling advanced NLP capabilities that can facilitate dynamic system interactions and real-time decision-making. This is particularly valuable in space-based and remote communication infrastructures, where autonomous operations are critical. By interpreting natural language inputs and commands, LLMs can enhance system responsiveness and streamline complex network management tasks, such as policy adaptation, fault detection, and resource optimization.

However, realizing the full potential of LLMs in MeNT-WiNs requires overcoming challenges. One of the critical

challenges in deploying LLMs within MeNT-WiNs lies in adapting these models to resource-constrained environments characterized by limited bandwidth, intermittent connectivity, and stringent energy constraints. Traditional LLMs, which are computationally intensive and rely on centralized, high-performance infrastructures, are unsuitable for such decentralized, resource-constrained environments. DRL algorithms can optimize model compression techniques, such as pruning and quantization [245], by exploring trade-offs between accuracy and resource usage. Furthermore, DRL can dynamically manage power consumption and task allocation, ensuring energy-efficient and real-time operation of LLMs across edge-based infrastructures prevalent in MeNT-WiNs. By leveraging DRL, the next generation of LLMs can be designed to be lightweight, energy-efficient, and capable of maintaining high performance and accuracy in constrained environments.

J. EE in DRL-Enabled MeNT-WiNs

The integration of DRL into MeNT-WiNs presents significant challenges related to energy consumption, particularly given the limited resources of platforms like AAVs, LEO satellites, and HAPs. DRL algorithms, especially those involving deep NNs, require substantial computational resources for both training and inference. The training phase, in particular, is highly energy-intensive, as DRL models must process large datasets and perform numerous iterations to update their parameters. Similarly, the inference phase, where DRL models make real-time decisions, can lead to high energy consumption, especially on platforms with constrained power budgets.

To address these energy challenges, several strategies can be implemented. Model compression techniques, such as pruning and quantization [245], can reduce the size and complexity of DRL models, thereby decreasing computational requirements and energy usage during both training and inference. Additionally, energy-aware resource allocation strategies can be employed, where DRL itself can be used to optimize the distribution of computational and communication tasks across the network, minimizing energy consumption while still meeting performance requirements. Another promising solution is offloading tasks from energy-constrained nodes (like AAVs) to more powerful edge servers or MEC nodes, which can process the data without draining the local platform's energy reserves. Furthermore, the incorporation of FL into DRL systems reduces the need for energy-intensive data transfers by ensuring that only model updates (rather than raw data) are shared between distributed nodes [264]. These techniques enable DRL to be more energy-efficient, making it viable for large-scale deployment in resource-constrained environments like MeNT-WiNs.

K. Securing DRL-Enhanced MeNT-WiNs

While DRL offers substantial benefits in resource management and system optimization, its integration into MeNT-WiNs introduces several new security concerns that must be addressed to ensure the robustness and reliability of these

networks. One of the primary risks is the vulnerability of DRL models to adversarial attacks [265], where malicious entities can manipulate the input data or training process to cause the model to make suboptimal or harmful decisions. For example, attackers could target the task offloading process in a satellite network by manipulating network traffic to force a DRL-based system into making inefficient resource allocation decisions, leading to network congestion or service degradation. Similarly, data poisoning attacks could compromise the learning process of DRL models by introducing erroneous data into the system, corrupting the trained model, and reducing the overall system reliability.

To mitigate these security threats, various solutions can be employed. Secure FL [266] is one such approach, where multiple agents collaboratively train DRL models without sharing sensitive raw data, thus preserving privacy while preventing malicious tampering. Additionally, anomaly detection systems powered by DRL [162] can continuously monitor network traffic for unusual patterns or behaviors that may indicate potential attacks, such as unauthorized access attempts or DDoS attacks. DRL can also be used to enhance adaptive security policies, dynamically adjusting encryption mechanisms, access controls, and intrusion detection protocols [267] in response to real-time network conditions or identified threats. This adaptive approach allows MeNT-WiNs to respond proactively to security challenges while maintaining high levels of performance. By integrating robust security mechanisms with DRL, MeNT-WiNs can ensure secure, efficient, and resilient operations, safeguarding sensitive data and preventing potential disruptions in non-terrestrial communication systems.

In summary, this section explored the intersection of DRL and advanced technologies within MeNT-WiNs. Topics such as semantic communication, quantum communications, ISAC, FL, DTs, LLMs, and mmWave integration were discussed in relation to how DRL can be leveraged for performance optimization. The section also addressed pressing research challenges, including scalability, security, heterogeneous integration, and energy efficiency. We identified several open research directions that merit further investigation for the practical realization of DRL-enhanced MeNT-WiNs.

V. CONCLUSION

This survey comprehensively explored the transformative integration of DRL with MEC, emphasizing its pivotal role in enabling intelligent decision-making in next-generation space-air-ground networks, collectively referred to as MeNT-WiNs. We reviewed the foundational principles of AI, ML, and DRL, specifically, we highlighted the classification of DRL into value-based, policy-based, and actor-critic methods, which serve as the basis for intelligent control in complex, dynamic environments. The survey systematically examined the application of DRL in various non-terrestrial platforms, including HAPs, satellites, and AAVs. These platforms face unique challenges such as intermittent connectivity, limited energy and computation capabilities, long propagation delays, and dynamic mobility. We discussed how DRL and MEC facilitate adaptive resource allocation, mobility-aware task

offloading, and intelligent service orchestration under such constraints. Furthermore, we illustrated how DRL can enhance scalability, EE, and latency reduction in MeNT-WiNs through case studies and recent works. This survey provided a detailed roadmap for future research, offering insights into the evolving landscape of DRL-assisted MeNT-WiNs and paving the way for further innovation in intelligent, space-based wireless communication systems. Addressing these challenges is essential for achieving robust, real-world deployment of DRL in MeNT-WiNs. Overall, this survey offered a structured roadmap for future research and innovation in DRL-assisted MeNT-WiNs. By bridging the gap between AI-driven decision-making and next-generation wireless infrastructure, it aimed to inspire the development of scalable, intelligent, and resilient communication systems across terrestrial and non-terrestrial domains.

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